Knowing When to Do Less: Adaptive AI Withdrawal in Children's Creative Drawing

ANONYMOUS AUTHOR(S)

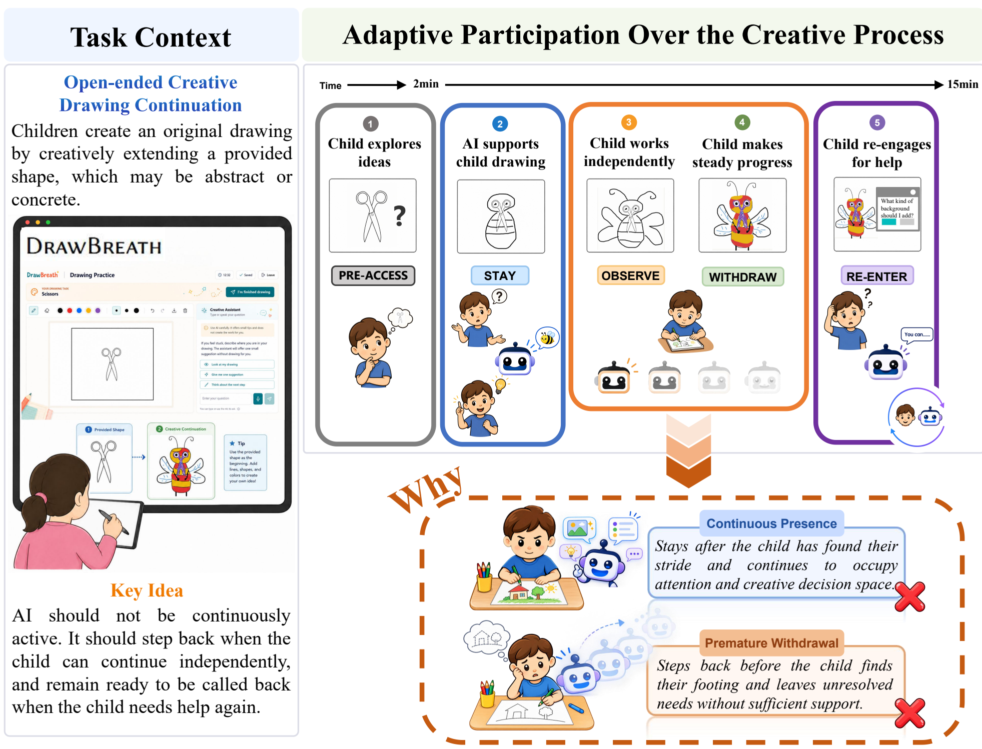

Fig. 1. DrawBreath regulates AI participation during an Open-ended Creative Drawing Continuation task. As a child develops a drawing from a provided shape, the system produces internal *Stay*, *Observe*, and *Withdraw* judgments; only an executable *Withdraw* judgment changes the interface, through a reversible fade. The design examines the tension between withdrawing before a support need is resolved and remaining salient after independent creation has resumed.

AI can support children's creative work, but its continued presence may outlast the need for help. We investigate adaptive withdrawal through DrawBreath, a drawing system that uses canvas activity and interaction history to determine when to stay present, observe, or withdraw, with interruptible fading and child-initiated re-entry. In a single-condition study, 65 sixth-grade children completed an open-ended drawing continuation task. We analyzed participation judgments, interaction logs, and post-task evaluations. Stable

Manuscript submitted to ACM

drawing followed 80% of completed withdrawals, and children generally reported being able to continue drawing after withdrawal, suggesting that withdrawal often occurred at moments compatible with continued independent creation. Illustrative trajectories situated withdrawal within ongoing elaboration, revision, and later reorientation, sometimes followed by renewed help-seeking. These findings frame withdrawal as a reversible adjustment within creation and inform the design of AI support that can step back while remaining accessible. Code, data, and study materials are available at https://draw-breath.github.io/.

CCS Concepts: • **Human-centered computing → Human computer interaction (HCI)**; *HCI theory, concepts and models*; *Interactive systems and tools.*

Additional Key Words and Phrases: adaptive withdrawal, human–AI co-creation, mixed-initiative interaction, AI participation, children's creativity, creative drawing, generative AI

ACM Reference Format:

Anonymous Author(s). 2027. Knowing When to Do Less: Adaptive AI Withdrawal in Children's Creative Drawing. In *Proceedings of CHI Conference on Human Factors in Computing Systems (CHI '27).* ACM, New York, NY, USA, 36 pages. https://doi.org/XXXXXXX.XXXXXXX

1 Introduction

Creative drawing gives children a way to develop and communicate ideas through visual marks [21]. As a drawing takes shape, children can explore an interpretation, revise it, and discover possibilities they had not initially imagined. AI tools can contribute to this process by helping children generate ideas, elaborate visual stories, and work through difficulties in creative projects [20, 23, 45, 71]. Supporting creation also means leaving children room to develop their own direction [25, 56, 60].

A child's need for support can change before the drawing is finished. A suggestion may help establish a direction, after which the child can continue independently; later revision may create a reason to seek help again. Scaffolding research emphasizes adjusting support to learners' changing needs and gradually transferring responsibility to them [50, 55, 63]. For an AI assistant, this raises a practical tension: stepping back too soon can remove advice that a child still needs, while remaining visible can keep assistance in the foreground after the child has moved on. The same question arises when a child makes progress without requesting help at all. Designing AI participation therefore involves deciding how long its presence remains useful, both during and between support exchanges.

Research on the timing of AI assistance provides a starting point. Studies of LLM access in writing ideation and critical thinking show that when assistance becomes available can shape its effects, with outcomes depending on the surrounding task and time available [54, 72]. Proactive systems also use ongoing activity to decide when to offer help, while confronting trade-offs between assistance and disruption [51]. These studies motivate adapting participation to the user's process, but leave open how an assistant should decide to recede from view once access has begun. In children's creative work, we know less about how AI can make this decision from the developing activity and how its withdrawal fits within the creative work that continues afterward.

We investigate this problem through an Open-ended Creative Drawing Continuation task adapted from the structured incomplete-shape drawing paradigm introduced by Barbot and later used in AuDrA [7, 49]. Children develop an original drawing by incorporating, extending, or reinterpreting a provided shape, without a prescribed final image. This setting makes the timing problem concrete: an unfinished drawing can already offer a direction for independent work, while further changes can bring new support needs. Yet visible progress alone does not establish whether advice is still useful. Studying withdrawal in this setting therefore requires examining both the assistant's participation decisions and the evolving creative trajectories in which they take effect.

In this work, we investigate **adaptive withdrawal as a way for AI to regulate its participation during children's creative drawing**. We define adaptive withdrawal as the process of using evidence from ongoing creation and interaction to decide whether to remain present, observe further, or step back while preserving the child's ability to restore support. We ask:

RQ1. Can AI autonomously regulate its participation by determining when to stay present, observe, or withdraw during an Open-ended Creative Drawing Continuation task with children based on the unfolding creative process?

RQ2. How is AI withdrawal situated within children's evolving visual creative processes during an Open-ended Creative Drawing Continuation task?

To address these questions, we designed DrawBreath, a drawing system with an AI assistant whose foreground presence adapts to canvas activity and interaction history (Figure 1). A decision agent produces internal *Stay*, *Observe*, and *Withdraw* judgments. *Stay* maintains availability when the evidence favors continued presence, whereas *Observe* keeps the same interface available while deferring withdrawal under uncertainty. An executable *Withdraw* judgment begins a gradual fade that the child can cancel. After withdrawal, children can restore the previous conversation or start a new question.

We conducted a single-condition study with 65 sixth-grade children completing one drawing task lasting up to 15 minutes. Each task began with two minutes of independent drawing before AI access opened. For RQ1, we examined model-authored judgments and enacted withdrawals alongside subsequent drawing, AI re-entry, and children's post-task evaluations. For RQ2, we located the first completed withdrawal within each child's task time and accumulated canvas activity, then examined time-ordered snapshots to understand the visual development surrounding it. This design provides a descriptive account of how adaptive withdrawal operated within children's creative work.

The system reduced its foreground presence while children generally continued creating. Only 11 children explicitly requested AI support, and most *Withdraw* judgments concerned unused AI presence, extending the empirical focus beyond stepping back after giving help. Following 65 completed withdrawals across 61 children, drawing was the first observed event within 90 seconds in 57 cases, and 52 met our criterion for stable continuation. Children were more decisive about being able to continue drawing than about whether withdrawal matched their support needs. First completed withdrawals generally occurred before the midpoint of both task time and accumulated activity. Illustrative visual trajectories showed subsequent elaboration, revision, and later reorientation accompanied by renewed help-seeking. Together, these findings characterize withdrawal as a reversible adjustment within ongoing creation, while leaving the comparative benefits and optimal timing of withdrawal for future investigation. Code, data, and study materials are available at https://draw-breath.github.io/.

Our work positions the regulation of continued AI presence as a design concern in human–AI creative collaboration. We make three contributions:

- A framing of *adaptive withdrawal* that treats stepping back as an intentional, process-contingent, and reversible form of AI participation.
- DrawBreath, a child–AI drawing system that operationalizes this framing through *Stay*, *Observe*, and *Withdraw* judgments, interruptible withdrawal, and child-initiated re-entry.
- An empirical account of autonomous participation regulation and withdrawal within children's evolving visual trajectories, informing the design and evaluation of AI support that can recede while remaining accessible.

2 Related Work

2.1 AI Participation and Creative Control in Human–AI Co-Creation

Human–AI co-creative systems organize participation through choices about initiative, turn taking, contribution, communication, and role allocation. COFI formalizes these as separable dimensions of interaction design and shows that co-creative systems can distribute initiative and contributions in substantially different ways [57]. Creative systems for children instantiate this design space through varied structures [23, 45]. StoryDrawer combines conversational and sketch-based contributions in visual storytelling [71]; Mathemyths alternates turns between a child and an AI storytelling partner [70]; and Words to Wonder introduces generated words and images at different points in children's story creation [24]. MusicScaffold instead assigns the AI stage-specific roles as guide, coach, and partner during adolescent music creation [32]. Together, these systems show that participation is configured not only through the content AI produces, but also through when it contributes and the role it occupies within the creative activity.

Creative control also depends on what people can do with an AI contribution. Wordcraft lets writers solicit and incorporate generated material within an editable writing environment [69]. Children using Scratch Copilot requested support across ideation, implementation, and debugging, but also adapted or rejected suggestions to preserve their project direction [20]. In co-drawing, users preferred to lead the creative direction while delegating bounded assistance to the AI [46]. Across these systems, control is exercised through invocation, prompting, acceptance, rejection, revision, or bounded role assignment [15, 16, 22, 35]. These mechanisms preserve human authorship by allowing people to determine when AI contributions enter and how far they shape the developing work. They therefore frame creative control as the coordination of participation rather than the maximization of AI contribution [5, 18, 19, 29, 73].

Taken together, this literature has largely framed creative control as the regulation of AI contributions: when AI acts, what it produces, how users incorporate or reject its output, and what role it plays in collaboration. However, participation is not limited to moments of contribution. An AI can influence the interaction simply through its continued presence, even without generating new content, while previously useful support may become unnecessary or remain unused as the creative process develops. This suggests that AI presence itself constitutes a dimension of participation that requires regulation. Creative control should therefore encompass not only what and how AI contributes, but also whether AI should remain present throughout the creative process.

2.2 Regulating AI Participation over Time: From Intervention to Withdrawal

Research on mixed-initiative and proactive systems establishes that AI action need not follow a fixed schedule. Mixed-initiative interfaces weigh the expected value of automated action against its potential costs [30, 48], while interruption-aware systems time notifications around task boundaries rather than delivering them indiscriminately [1, 33]. More recent systems infer intervention timing from ongoing behavior and context. Time2Stop adapts just-in-time interventions using sensed behavior and user feedback [47]; Codellaborator initiates programming support from editor activity and task context [51]; and proactive assistance in collaborative story writing varies AI initiative while examining its effects on agency and creative contribution [67]. Together, this work supports process-contingent timing: whether the AI should act can depend on the user's current activity, task state, and interaction history. Such timing concerns extend beyond when AI should enter or act to how long it should remain involved.

Intervention timing nonetheless leaves a different temporal question unresolved. Studies of processing delays and access timing examine when an AI response appears or when AI support becomes available [39, 54]. Proactive systems likewise decide when to offer a message, suggestion, or generated artifact. These approaches primarily ask, *When* *should the AI act?* They do not necessarily ask, *When should the AI cease to occupy the interactional foreground?* (*AI withdrawal*). Silence is not withdrawal: producing no new content at a given moment does not indicate that the system has reassessed whether its continued presence remains useful. Nor is withdrawal equivalent to terminating support: stepping back can be designed as a change in participation state, with access remaining flexible and reversible.

One context for adjusting AI participation follows directly from scaffolding. Effective scaffolding is contingent on a learner's changing needs and ultimately transfers responsibility back to the learner [6, 44, 53, 63, 65]. In a *post-support withdrawal*, the AI has contributed to a support episode, the child has regained enough direction to proceed, and continued foreground assistance may no longer be needed. Fading in this case is not a timeout after help. It is a transition from active support toward reduced foreground involvement, guided by evidence that the support episode has run its course.

A second participation-adjustment context arises when no support episode occurs. Work on interruptions and proactive AI shows that assistance can impose attentional and workflow costs and can alter perceived control even when it is intended to help [33, 51, 67]. In an *unused-presence withdrawal*, the AI is available but the child continues independently without using it. Scaffolding has not been completed because it never began; instead, the marginal value of keeping the AI in the foreground may have become low. Post-support and unused-presence withdrawal thus follow different mechanisms, but both ask whether continued AI foreground participation is still warranted.

Mixed initiative is commonly expressed through additional system action. Yet the same cost–benefit logic also supports deliberate restraint: an AI can exercise initiative by reducing foreground participation when further presence is unlikely to help, while preserving a path for re-entry. Adaptive withdrawal thus frames reduced presence as an intentional and reversible participation-state transition, not as the termination or absence of support.

2.3 Creative Processes as Context for Adaptive AI Withdrawal

Creative activity unfolds as a trajectory that cannot be fully reconstructed from the final artifact. Drawing externalizes and transforms mental representations while also shaping what the creator notices and develops [21, 26, 64]. Studies of art making describe creation as recursive: artists generate, externalize, evaluate, revise, and sometimes redirect emerging work rather than proceed through a fixed sequence [10, 11, 41, 42]. Design research similarly characterizes creative work as a co-evolution of problem and solution spaces [17], and temporal studies distinguish linear from iterative process configurations [61]. Together, these accounts position idea emergence, exploration, revision, and stabilization as processes whose order and duration may vary across creators.

The Open-ended Creative Drawing Continuation task makes this distinction concrete. It builds on the structured incomplete-shape drawing paradigm introduced by Barbot and later used in AuDrA [7, 49]. AuDrA demonstrates how completed outputs from this paradigm can be evaluated automatically at the artifact level. Such assessment can characterize the completed drawing [3, 59], but it does not recover when an idea stabilized, when revision changed its direction, or when external support ceased to be useful. Trajectory evidence therefore serves two related purposes for adaptive withdrawal: it can inform whether the AI should remain involved, and it can locate a withdrawal transition within the child's broader creative process.

Observable process signals are informative but do not reveal intent on their own. Drawing behavior alternates between mark making and pauses, with its temporal organization varying across age groups [9, 43]; in a field study of art making, pauses supported planning and monitoring at different points in the drawing process [68]. Revision is similarly multivalent: it may refine, redirect, or test an emerging idea, and different distributions of revision can occur within trajectories that yield comparably creative outcomes [36, 61]. A pause, revision, or brief decline in canvas activity therefore records an observable event but cannot by itself establish whether the child is productively reflecting, encountering difficulty, completing the work, or disengaging. Its interpretation depends on preceding and subsequent activity and on convergence with other evidence. This ambiguity argues against triggering withdrawal from a single inactivity threshold or task-wide timer. A participation decision should instead relate recent drawing and revision activity to canvas progress, interaction history, the status of any support episode, prior participation states, and temporal context. When the available evidence remains insufficient or conflicting, an intermediate *Observe* judgment can defer withdrawal while preserving the possibility of later reassessment.

Process evidence also provides context after a withdrawal decision has been enacted. A withdrawal transition may occur while an interpretation is still forming, during independent elaboration, or amid revision and refinement. Examining where this transition occurs within a creative trajectory complements the question of how it was decided without treating trajectory position as proof that the decision was correct. Adaptive withdrawal therefore requires both participation regulation and trajectory-level contextualization: whether AI can determine when to *Stay*, *Observe*, or *Withdraw*, and how completed withdrawal is situated within heterogeneous creative processes.

3 DrawBreath

3.1 Designing and Operationalizing Adaptive Withdrawal

3.1.1 System Overview and Interaction Design. DrawBreath was implemented as a two-pane interface comprising a drawing workspace and an AI sidebar (Figure 2B). The drawing workspace included a canvas and a toolbar with colored pens, an eraser, and adjustable stroke widths. Whenever AI support was available, the sidebar displayed the child–AI exchange as a sequence of messages and allowed children to select a predefined question or enter a free-text request. Participation regulation operated through the sidebar, leaving the drawing workspace and its controls unchanged.

The sidebar was governed by three internal judgments, *Stay*, *Observe*, and *Withdraw*, whose labels were not shown to children. *Stay* and *Observe* shared the same available interface configuration; only an executable *Withdraw* judgment initiated a cancellable fade. Independently of these judgments, messages received a translucent mask when the pointer left the dialogue region and became legible again when the pointer returned. Figure 2C summarizes the sidebar's progression from the pre-access period through availability and active support to gradual withdrawal, completed withdrawal, and child-initiated re-entry. Sections 3.1.3–3.1.6 describe the judgment and execution mechanisms.

3.1.2 Deriving Design Principles for Adaptive Withdrawal. We translated prior work on scaffolding, mixed-initiative interaction, and context-aware assistance into three design principles for the participation policy.

DP1: Preserve an initial period for self-directed interpretation. The temporal placement of AI access can change how people formulate ideas and engage with a task [39, 54, 72]. We therefore introduced an AI-free opening period in which children could begin interpreting the provided shape before AI contributions became available. Its duration was fixed through the formative calibration in Section 3.2.

DP2: Base withdrawal on the unfolding process, not on a single timeout. Scaffolding frames fading as a contingent transfer of responsibility rather than the expiration of a preset interval [63, 65], while context-aware systems infer intervention timing from behavior and interaction history [47, 51, 52]. Accordingly, inactivity alone could not trigger withdrawal. Drawing studies show that pauses can serve planning and monitoring, while revision is distributed across heterogeneous creative trajectories [36, 61, 68]. Because these observable events do not reveal a child's reason for acting or pausing, the Decision Agent interpreted them jointly with canvas progress and interaction history.

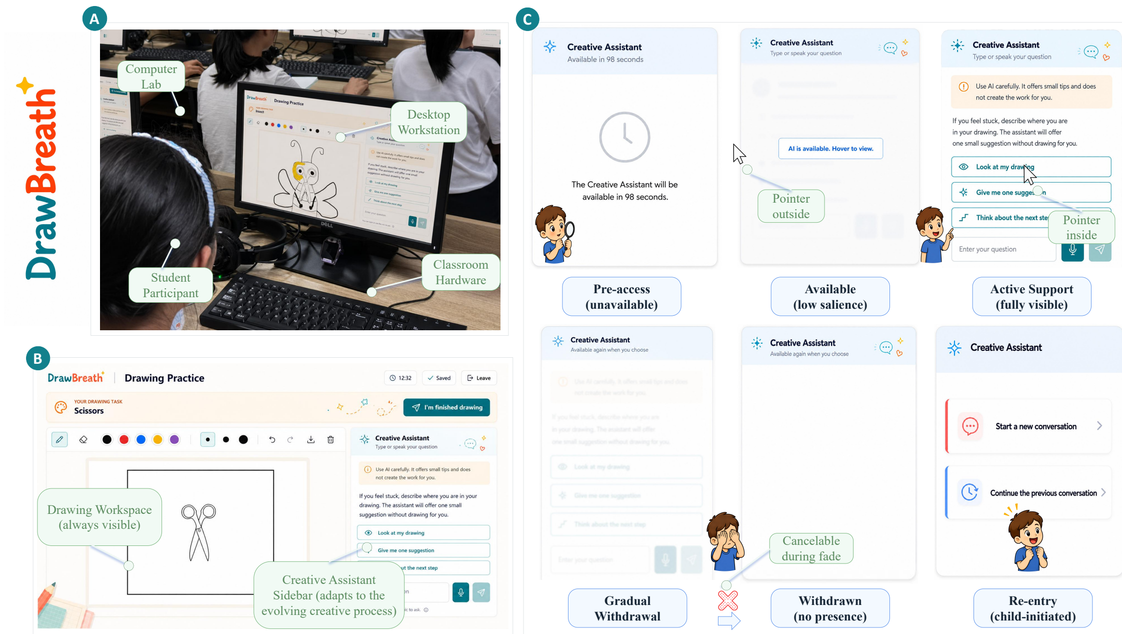

Fig. 2. System setting, interface, and adaptive sidebar behavior in DrawBreath. (A) A study session in the school's computer lab. (B) The two-pane interface, comprising a persistent drawing workspace and an AI assistant sidebar. (C) The sidebar progression from pre-access through low-salience availability and active support to gradual withdrawal, completed withdrawal, and child-initiated re-entry; the fade could be cancelled before withdrawal completed.

DP3: Treat adaptive withdrawal as a staged, interruptible, and reversible process. Mixed-initiative systems must weigh the expected benefit of acting against the costs imposed on the user [30, 48], while proactive assistance can be useful yet disruptive when poorly timed [51, 67]. Human–AI interaction guidelines further emphasize user control and opportunities to correct system behavior [4], and co-creation frameworks distinguish system initiative from user authority [57]. We therefore avoided treating withdrawal as a single, irreversible decision. When process evidence remained conflicting or insufficient, *Observe* deferred withdrawal without changing the interface or introducing a new contribution. When the evidence supported stepping back, only a validated *Withdraw* judgment could initiate a progressive fade; renewed access cancelled the transition, and child-initiated re-entry restored AI availability after completion. Thus, the system could reduce its foreground presence while the child retained authority to interrupt or reverse the process.

3.1.3 *Process Evidence for Participation Regulation.*

The timing and sequence of creative activity preserve information that a finished image does not [7, 21, 43, 61]. DrawBreath therefore organized an initial pool of candidate evidence into three groups: canvas activity, AI engagement, and temporal and data-quality context. Event-level measures and snapshots were summarized over recent windows and the session history, then interpreted jointly. No single event was treated as evidence of a child's internal state or as a withdrawal trigger.

Canvas activity and revision. Time-stamped canvas events and successive snapshots summarized recent production and cumulative visual change, including activity level, spatial and coverage change, pauses, revision, and tool use. These system-defined measures described observable activity rather than latent intent. Consistent with DP2, the Decision Agent interpreted pauses, revision, and low activity in relation to surrounding canvas and interaction evidence rather than as standalone indicators.

AI engagement and support episodes. Consistent with contingent scaffolding's focus on the learner's response to support [63], we reconstructed support episodes from sidebar access and request submission through response delivery, return to the canvas, and possible re-engagement. When no support episode occurred, the absence of AI use was considered only alongside canvas activity and could not by itself justify withdrawal. The policy considered whether an exchange remained active, whether drawing resumed after a response, and whether the child quickly sought further support. Pointer presence was retained only as interface activity, not as evidence that a message had been read or understood.

Temporal context and evidence quality. Each evaluation combined recent activity with cumulative history, task time, prior participation judgments, interrupted withdrawals, re-entry, and flags for missing or stale data [47, 52]. Simple baselines summarized recent canvas progress, repeated revision, and unresolved support episodes, but did not assign a participation judgment or trigger an interface transition. Missing activity or incomplete evidence therefore could not by itself support withdrawal. Section 3.2 describes how this candidate pool and its baselines were calibrated.

3.1.4 Decision Agent for Participation Regulation. At each review opportunity, the process evidence described above was assembled into an evidence packet and passed to the Decision Agent. Both the student-facing support component and the Decision Agent used GPT-5.5 through the OpenAI Chat Completions API. Decision calls used a temperature of 0.2 and returned structured JSON. The agent synthesized the evidence into a *Stay*, *Observe*, or *Withdraw* judgment, which the runtime handled separately from the interface enactment described below. The complete model settings, decision prompt, evidence-packet and return schemas, and response-validation contract are provided in Appendices C and D. Source code and further implementation details are available at https://draw-breath.github.io/.

3.1.5 Stay–Observe–Withdraw Decision Architecture. The architecture separated three functions: judgment, runtime control, and interface enactment. First, the Decision Agent produced a model-authored *Stay*, *Observe*, or *Withdraw* judgment, indicating whether the AI should remain available, defer withdrawal while awaiting further evidence, or begin stepping back. Second, the runtime applied that judgment under the system's interaction safeguards and could maintain a temporary hold when a model review or transition was not currently permitted. These holds were logged separately from model-authored judgments. Third, the interface enacted the resulting assignment as a visible, fading, or withdrawn sidebar. The three layers did not map one-to-one: *Stay* and *Observe* expressed different judgments but both kept the sidebar visible, whereas *Withdraw* initiated only a cancellable fade. The sidebar became withdrawn only if that fade completed without child interruption.

Table 1 summarizes how the agent interpreted the three participation judgments and how each judgment could be handled by the runtime and interface.

Table 1 specifies evidential semantics rather than fixed rules; the agent interpreted the sequence and context jointly. For *Withdraw*, we distinguished *unused-presence withdrawal*, when the child showed credible independent progress without an active support episode, from *post-support withdrawal*, when the child had obtained enough support to resume drawing. The latter allowed partial resolution when subsequent canvas activity indicated that the child could act on the support. The support-episode label *not used* did not by itself establish that the child had ignored the available AI.

Table 1. Operational semantics of the three model-authored participation judgments in DrawBreath.

Decision	Selection evidence	Participation response	Subsequent handling
Stay	Evidence favors keeping the AI immediately available: the child may be engaged in an active support episode or repeatedly returning to the dialogue, or the unfolding process may provide substantive evidence that receding could be premature.	Preserve any ongoing support episode and keep the dialogue immediately available without generating unsolicited content.	Re-evaluate when the interaction or process evidence changes, or at the next scheduled review.
Observe	Evidence remains substantively inconclusive: indicators of independent progress and possible support need conflict, the child's next action is not yet interpretable, or the available evidence is too limited or unreliable to support withdrawal.	Defer withdrawal, keep the sidebar available in the same interface configuration as *Stay*, and continue collecting canvas and interaction evidence without generating a new message.	Review again after a short interval or when a new event helps resolve the uncertainty.
Withdraw	Evidence favors stepping back: continued foreground presence appears to have low marginal value, the child shows credible independent progress, and no active exchange or blocking support need would be interrupted.	Propose a withdrawal transition; if the runtime safeguards pass, begin a progressive and interruptible fade rather than disappearing immediately.	Complete withdrawal if the child does not re-engage; otherwise, cancel the transition and restore the exchange.

The runtime layer produced separate non-model assignments. When AI access first opened after the two-minute AI-free period, the runtime initialized an *Observe* hold to indicate that the sidebar was available while additional process evidence accumulated. Runtime safeguards could also maintain an *Observe* hold when a model review was deferred, for example during active child input or temporary model unavailability. After child-initiated re-entry, the runtime immediately restored the sidebar and assigned *Stay*, because the child's action itself established a current request for AI availability. These runtime records were logged separately and were not included among model-authored participation judgments.

The Decision Agent's output was treated as a proposed participation judgment rather than a direct interface command [28, 30, 48, 66]. The runtime normally enacted a valid judgment, but could defer or withhold its execution when it conflicted with an operation already in progress. Such conflicts were rare and were logged together with the evidence packet, model-authored judgment, and eventual runtime outcome. This separated the agent's decision from what the interface ultimately enacted. We use *withdrawal process* for the sequence from a *Withdraw* judgment through the fade to either completed or cancelled withdrawal.

3.1.6 Reversible Withdrawal and Re-entry. Withdrawal was implemented as an interruptible proposal to step back rather than an irreversible system verdict. Once a current and executable *Withdraw* judgment had passed the safeguards above, the sidebar entered a brief progressive fade. The interface did not tell the child that support was no longer needed, and the canvas, toolbar, and available drawing area remained unchanged. During the fade, renewed access to the dialogue region immediately cancelled the transition and restored the ongoing exchange. Cancellation was handled locally so that the child did not have to wait for another model judgment, while the attempted withdrawal and its cancellation were retained in the interaction record.

If the fade completed, the preceding messages and direct-input controls left the interactional foreground. Hovering over the sidebar then exposed two re-entry choices without automatically reopening the conversation. *Resume previous conversation* restored the preceding exchange and its support context, offered a concise reminder of the earlier issue, and allowed the child to revisit or clarify unresolved support without receiving additional suggestions by default. *Start a new question* opened a new support episode while retaining the current canvas context, but did not assume that the earlier issue remained relevant. Either action immediately restored the sidebar and produced a runtime-assigned *Stay*, because child-initiated re-entry itself established that the AI should remain available. This assignment did not require another model judgment and was logged separately from model-authored *Stay* decisions. Both actions returned control of AI access to the child, while distinguishing continuation of an earlier need from the emergence of a new one [4, 57].

The logs distinguished returning to the preceding conversation from starting a new question. Either action provided process evidence for evaluating withdrawal timing, but neither was treated as a standalone ground-truth label: a rapid return could suggest premature withdrawal, whereas a new question after continued drawing could mark a later creative phase.

3.2 Formative Development and Calibration

Before formal data collection, we conducted a series of instrumented system trials and a separate small-scale formative pilot with children from the target school stage to calibrate the process-evidence representation and establish the timing parameters governing how AI participation was introduced, evaluated, and withdrawn. Children in the formative pilot completed the full task flow. We then reviewed their time-stamped canvas traces, snapshots, AI interactions, model-authored judgments, runtime assignments, and withdrawal execution outcomes, together with brief checks of how they understood the interface behavior, re-entry choices, and questionnaire language. These trials were used to settle the study configuration and identify implementation problems, not to estimate the study outcomes; the resulting records were not included in the formal dataset.

The 2-minute AI-free opening and the 15-minute task duration were selected from these timing trials rather than set solely by design intuition. We inspected children's canvas states at candidate checkpoints after task onset. The 2-minute point was retained because children generally had enough time to inspect the starting shape and either begin drawing or establish an initial direction, while their work remained sufficiently open for subsequent support to be meaningful. The 15-minute duration was retained because it allowed children to develop an assessable drawing and, after the opening period, left enough time for voluntary AI use, observation, withdrawal, and re-entry to unfold. The opening period counted toward the 15 minutes. Once both timings met these practical criteria in the formative trials, they were fixed before the formal study and were not adjusted in response to the formal data.

The same trials were used to reduce and stabilize the broad candidate pool described in Section 3.1.3. At each candidate decision point, we compared the evidence packet available to the AI with a replay of the corresponding event sequence and canvas snapshots. A measure was retained for real-time judgment only when it was available at the decision point, could be reconstructed consistently, added information beyond closely related summaries, and could be interpreted without assuming an unobserved mental state. Measures that were redundant, too sparse, overly sensitive to network or snapshot delay, or meaningful only through unsupported inferences about attention, understanding, or difficulty were removed from the live packet, although their raw events remained available for later process analysis. Through repeated debugging and replay, we fixed the real-time evidence set around canvas advancement, pauses and revision, current AI dialogue activity and short-interval returns to the dialogue, support-episode status, resumption of self-directed drawing, task time, and data quality. We likewise adjusted the summary windows and simple baselines until they produced stable, interpretable descriptions of the unfolding sequence; they summarized evidence but never mechanically assigned a model-authored participation judgment.

We calibrated the decision prompt and response schema against boundary cases among the three judgments and tested incomplete or stale evidence, concurrent child input, ongoing AI generation, invalid responses, and fade cancellation. The same replay trials were used to fix the Decision Agent's temperature at 0.2 before formal data collection. Separate interaction trials fixed a 450-ms hover delay, a 650-ms leave delay, and a 2.5-second cancellable fade. The trials also fixed a 15-second minimum-availability safeguard after AI access opened. During this interval, runtime maintained an *Observe* hold and did not permit the model to consider withdrawal; the safeguard did not trigger withdrawal or determine a later judgment. The agent received a review opportunity every 30 seconds or after a meaningful event. Evidence summaries covered approximately 8–15 seconds of immediate activity and 20–30 seconds of short-term development, with 60 seconds retained as broader context. After testing, we froze the evidence set, summary windows, prompts, response schema, safeguards, interface timings, and logging configuration for all formal sessions.

4 Method

4.1 Study Design

4.1.1 Participants. We recruited 72 sixth-grade students from a public primary school. Seven students were excluded because they did not complete the study session or provided blank submissions, resulting in a final sample of 65 participants. The sample was approximately balanced by gender. Participation was voluntary. The study protocol received ethics approval from the authors' institution. We obtained informed consent from the children's legal guardians and assent from the children before participation.

4.1.2 Open-ended Creative Drawing Continuation Task. Each participant completed one 15-minute Open-ended Creative Drawing Continuation task adapted from the structured incomplete-shape drawing paradigm introduced by Barbot and later used in AuDrA [7, 49]. Each child was randomly assigned a single starting shape, which could be abstract or representational and appeared on an otherwise blank canvas. Children were asked to develop an original drawing by incorporating, extending, or reinterpreting the provided shape; the task specified neither a target image nor a single correct completion.

This bounded but open-ended structure allowed children to follow different creative paths, providing the process variation needed to examine changing support needs.

4.1.3 Study Procedure. Each participant took part in one study session in the school's computer lab using a desktop workstation (Figure 2A). Children were told that the AI might become less visually prominent or temporarily step back and that they could restore it through the sidebar, but not when this might occur or which process signals informed it. This limited disclosure made withdrawal understandable without cueing children to anticipate its timing.

The countdown began when the task interface opened. During the first two minutes, which counted toward the total task time, the AI sidebar remained empty and unavailable while children inspected the assigned shape and considered possible directions. AI support then became available, and runtime initialized an *Observe* hold for the calibrated 15-second minimum-availability interval. Thereafter, the participation architecture described in Sections 3.1.3–3.1.6 operated for the remainder of the task.

Participants could submit their drawing at any point before the deadline; otherwise, the system submitted it automatically when the countdown elapsed. After submission, participants completed a post-task questionnaire. The questionnaire measures and interaction records used in the analyses are described in Section 4.2.

4.2 Data Sources and Measures

The dataset combined time-stamped system records with a post-task questionnaire. Each participant contributed one task record, but a record could contain multiple model-authored participation judgments and multiple withdrawal processes. We therefore retained the task record, participation judgment, and completed withdrawal as distinct log-derived units. All sources were linked through participant, session, decision, and withdrawal identifiers and ordered on the formal-task timeline.

4.2.1 Drawing and Interaction Event Logs. During each formal task, DrawBreath recorded the canvas and AI-interaction events described in Section 3.1.3, together with canvas snapshots, task boundaries, and the session configuration in effect. Canvas records captured strokes, erasures, undo and redo operations, and tool changes. AI-interaction records captured sidebar access, predefined and free-text requests, message generation and completion, and the re-entry actions described in Section 3.1.6. Sequence numbers and task-relative timestamps were used to reconstruct the order of events within each task record.

For event-level analyses, a drawing action was defined as the initiation of a stroke or erasure. Explicit AI help-seeking required submission of a predefined or free-text question; opening or hovering over the sidebar alone was retained as observable interface activity and was not interpreted as evidence that a child had read, understood, or used an AI response. Task-level summaries described the amount and temporal distribution of drawing, revision, inactivity, and AI interaction. Measures used to position and contextualize withdrawal within creative trajectories were derived from these event records as specified in Section 4.3.2.

4.2.2 Participation Judgments and Withdrawal Events. We treated participation judgments and withdrawal events as linked but distinct log units. A judgment entered the model-authored dataset only when the Decision Agent returned a valid *Stay*, *Observe*, or *Withdraw* response without a fallback or response error. Each record retained its task-relative time and order, structured support-need and support-episode fields, the withdrawal-pathway label where applicable, and the validation and execution outcome. Runtime holds and re-entry assignments were recorded separately and excluded from model-authored judgments.

For each model-authored *Withdraw* judgment, the runtime logs indicated whether a fade was initiated and, if so, whether it completed or was cancelled by child re-engagement. The judgment was also labeled as an unused-presence or post-support withdrawal according to Section 3.1.5. A task record could contain more than one completed withdrawal when re-entry was followed by a later withdrawal. Only completed withdrawals entered the post-withdrawal analysis.

Each completed withdrawal was aligned to fade completion ($t = 0$). Within the following 90 s, we identified the earliest drawing action, child-initiated AI re-entry, or task end; windows containing none of these were coded as no observed event. Re-entry included either resuming the previous conversation or starting a new question, and drawing after re-entry was not counted as independent post-withdrawal continuation. We counted stroke and erasure initiations until the earliest of re-entry, task end, or the end of the 90-second window.

4.2.3 Post-task Questionnaire. After submitting the drawing, each child rated six study-specific statements about DrawBreath's withdrawal behavior. Q1–Q3 addressed whether withdrawal occurred after independent continuation was possible, whether it occurred while previous advice was still needed (negatively worded), and its overall appropriateness. Q4–Q6 addressed interruption and two aspects of post-withdrawal continuity: knowing what to draw next and continuing with one's own ideas. The child-facing wording was checked during the formative work described in Section 3.2 and fixed before formal data collection.

Responses used a five-point scale: *Strongly disagree*, *Somewhat disagree*, *Uncertain*, *Somewhat agree*, and *Strongly agree*. Because the questionnaire was administered once after the task rather than after each withdrawal, responses were analyzed at the child/task-record level as overall evaluations, not as ratings of individual withdrawal episodes. We retained item-level distributions and did not compute a composite score because the items represented distinct facets of the experience. For the diverging visualization only, Q2 was directionally recoded and labeled by its positively keyed interpretation; textual reporting retains the meaning of the original item.

4.3 Analysis

Our analyses were descriptive and sequence-based. For RQ1, we examined model-authored judgments, withdrawal execution, post-withdrawal behavior, and children's post-task evaluations to characterize how the AI regulated its participation. For RQ2, we positioned withdrawal relative to task time and accumulated canvas activity, then examined time-ordered canvas snapshots to characterize the creative contexts surrounding withdrawal. Given the single-condition deployment and modest sample, the analyses describe how participation regulation operated and where withdrawal occurred within children's creative processes rather than estimate causal effects. Evidence sources remained analytically distinct and were not combined into a ground-truth measure of withdrawal correctness.

4.3.1 RQ1: Evaluating Adaptive Participation Decisions. We first reconstructed each task's enacted participation trajectory. For Figure 3, the 15-minute timeline was divided into 30-second intervals, each assigned the interface state occupying the largest share of that interval. AI availability, child-initiated re-entry, cancelled fades, and task end were overlaid as event markers. This visualization represented enacted interface states. Judgment-level analyses instead used the original timestamps of the valid model-authored outputs defined in Section 4.2.2.

At the judgment level, we summarized the frequencies of ***Stay***, *Observe*, and *Withdraw* and measured the interval from AI availability to the first *Withdraw* judgment in each record. Withdrawal was ineligible during the initial 15-second minimum-availability safeguard, which was treated as a runtime hold rather than a model-authored judgment. For presentation, first-withdrawal latencies were grouped into 15–30, 30–60, 60–120, and more than 120 s. We coded a first *Withdraw* as *direct* when it was also the record's first model-authored judgment and as *deferred* when one or more ***Stay*** or *Observe* judgments preceded it. For deferred cases, we also recorded the immediately preceding judgment. We summarized unused-presence and post-support withdrawals and compared the structured support-need, support-episode, and withdrawal-pathway fields with their predefined semantics. Because these fields were produced from the same evidence as the participation judgment, this comparison assessed internal policy alignment rather than independent timing accuracy.

Completed withdrawals were the unit of analysis for post-withdrawal behavior. Within the 90-second window defined in Section 4.2.2, we classified the earliest outcome as drawing, child-initiated AI re-entry, task end, or no observed event. We calculated latency to the first drawing action and the cumulative proportion of withdrawals followed first by drawing within 30, 60, and 90 s.

We further distinguished stable from fragmented drawing-first episodes using the number and temporal span of post-withdrawal drawing actions. The formal criterion is given alongside the corresponding results in Section 5.2.2.

These measures characterize whether creative activity continued after withdrawal. They do not establish an ideal withdrawal moment.

Finally, we summarized Q1–Q6 as item-level five-category distributions and reported combined agreement or disagreement percentages where substantively relevant. Across the RQ1 analyses, categorical outcomes were reported as counts and percentages, and timing measures as medians and interquartile ranges. We interpreted the task-level questionnaire responses alongside the event analysis while preserving their different units: the logs described behavior after individual completed withdrawals, whereas the questionnaire captured each child's overall withdrawal experience.

4.3.2 RQ2: Positioning and Contextualizing AI Withdrawal within Children's Creative Trajectories. The task record was the unit of analysis for RQ2. We selected the first completed withdrawal in each of the 61 records containing at least one completed withdrawal, yielding 61 task-level anchors and preventing the three records with repeated withdrawals from receiving additional weight. For each anchor, we calculated two normalized positions: a temporal position, defined as the task-relative withdrawal-completion time divided by the child's actual task duration, and an action-based position, defined as the canvas-operation sequence value at withdrawal completion divided by the final sequence value at task completion. Actual task duration ended at early submission or automatic submission at the 15-minute limit. The action-based measure accumulated the canvas events described in Section 4.2.1; it indexed drawing, revision, and tool-setting activity rather than visual completeness. For three records without a snapshot at withdrawal completion, the withdrawal sequence value was recovered from the time-stamped operation log.

We summarized both position measures using medians, interquartile ranges, and the number of records below the 0.5 midpoint. Their within-record relationship was examined by plotting paired positions against the equality diagonal and subtracting action-based position from temporal position. These comparisons were descriptive and were not interpreted as estimates of an optimal withdrawal point.

To contextualize the normalized positions, we examined time-ordered canvas snapshots together with the aligned event logs before and after each selected withdrawal. We compared whether a recognizable visual direction was present at withdrawal, whether subsequent changes elaborated an established direction, revised its visual realization, or reoriented the work, and whether later help-seeking and AI re-entry occurred. Figure 6 presents three contrasting records selected to make these process differences visible; they are illustrative cases rather than exhaustive categories or estimates of their prevalence. Visual completeness, direction stability, and later re-entry were used to describe the surrounding creative trajectory, not as ground truth that a withdrawal was correctly timed.

5 Results

5.1 Overview of the Analyzed Task Records

The final dataset comprised 65 valid task records, one per child. Each record included a submitted drawing and traceable canvas activity. Within the 15-minute task window, children recorded a median active time of 9.2 minutes (IQR = 7.5–11.1) and produced a median of 72 drawing strokes (IQR = 53–100), indicating sustained engagement with the drawing task and substantial work on the canvas.

Explicit AI help-seeking, by contrast, was concentrated among a small subset of participants. Only 11 of 65 children (16.9%) submitted at least one question to DrawBreath, whereas the remaining 54 (83.1%) completed the task without explicitly requesting AI support. Across the cohort, substantial child-led drawing activity therefore coexisted with relatively sparse and unevenly distributed explicit AI use.

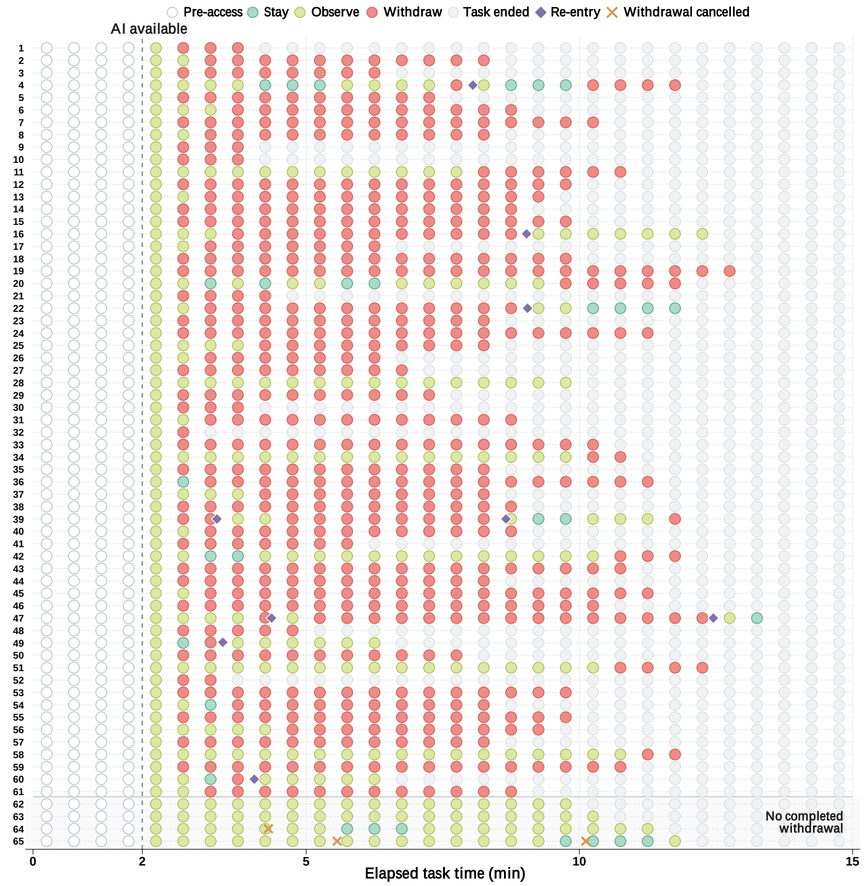

Fig. 3. AI participation-state trajectories across the 65 analyzed task records. Each row represents one child, and each circle shows the participation state that occupied the largest share of a 30-second interval. The vertical dashed line marks when AI support became available after the two-minute pre-access period. Colors distinguish pre-access, *Stay*, *Observe*, *Withdraw* (after a completed withdrawal), and task ended. Purple diamonds mark child-initiated re-entry, and orange crosses mark cancelled withdrawal transitions. Records below the horizontal divider had no completed withdrawal.

5.2 Evaluating DrawBreath's Autonomous *Stay*, *Observe*, and *Withdraw* Judgments

To address RQ1, Figure 3 first provides a session-level overview of how AI participation unfolded across the 65 task records. We then examine the model-authored judgment patterns themselves, including when the first *Withdraw* judgment emerged and whether it occurred directly or only after earlier *Stay* or *Observe* judgments. We next turn to evidence that became available only after withdrawal: children's subsequent creative activity and their post-task evaluations of the withdrawal experience. These sources are treated as complementary evidence about how the participation policy operated and was experienced, rather than as a single ground-truth measure of withdrawal correctness.

5.2.1 Temporal Emergence and Decision Pathways of Model-Authored Participation Judgments. Across the 65 task records, the decision agent produced 267 valid model-authored participation judgments: 59 *Stay* judgments (22.1%),

Table 2. Descriptive summary of model-authored participation judgments, timing of the first *Withdraw* judgment, and withdrawal contexts.

Analysis level	Pattern	Count	Percentage or summary
All model-authored judgments	*Stay*	59	22.1%
	Observe	131	49.1%
	Withdraw	77	28.8%
Timing of first *Withdraw*	15–30 s	34	54.0%
	30–60 s	16	25.4%
	60–120 s	7	11.1%
	After 120 s	6	9.5%
	Median time	–	29 s (IQR = 19.5–50.5 s)
All model-authored *Withdraw* **judgments**	Unused-presence	59	76.6%
	Post-support	18	23.4%

Note. Denominators are all valid model-authored judgments (N = 267), records with a first *Withdraw* judgment (n = 63), or all model-authored *Withdraw* judgments (N = 77), as appropriate. Timing begins when AI access became available; withdrawal was ineligible during the first 15 s.

131 *Observe* judgments (49.1%), and 77 *Withdraw* judgments (28.8%; Table 2). These counts exclude runtime-generated holds and re-entry assignments, including the initial *Observe* hold shown in Figure 3.

At least one model-authored *Withdraw* judgment occurred in 63 records (96.9%). First *Withdraw* judgments were concentrated early in the AI-available period: 34 of 63 (54.0%) occurred within 30 s and 50 (79.4%) within 60 s. The median was 29 s (IQR = 19.5–50.5 s).

Among the 63 records with a first *Withdraw* judgment, 43 (68.3%) followed a direct pathway, in which *Withdraw* was the first valid model-authored judgment. The remaining 20 (31.7%) followed a deferred pathway, with one or more *Stay* or *Observe* judgments occurring first. *Observe* immediately preceded withdrawal in 16 of these 20 deferred records (80.0%), while *Stay* preceded it in four (20.0%). Thus, although withdrawal often appeared as the agent's first valid judgment, nearly one-third of records reached it only after earlier participation judgments, most commonly through *Observe*.

Of the 77 *Withdraw* judgments, 59 (76.6%) were unused-presence withdrawals and 18 (23.4%) were post-support withdrawals. The structured fields also showed internal policy alignment: 74 of 77 *Withdraw* judgments classified current support need as low, while 121 of 131 *Observe* judgments classified it as uncertain and 10 as medium. As specified in Section 4.3.1, this alignment does not independently assess withdrawal timing. We therefore turn to what children did after the AI stepped back.

5.2.2 Children's Creative Activity Following Completed Withdrawal. Of the 77 model-authored *Withdraw* judgments reported above, 68 initiated a logged withdrawal process. Sixty-five of these processes reached completion, whereas three were cancelled during the interruptible fade. The 65 completed withdrawals occurred across 61 task records; because re-entry could be followed by a later withdrawal, three records contributed more than one completed withdrawal.

Drawing was the first observed event in 57 of the 65 completed withdrawals (87.7%). Among these cases, the median latency to the first drawing action was 5.6 s (IQR = 2.6–18.9 s). Drawing occurred first after 49 completed withdrawals (75.4%) within 30 s, 55 (84.6%) within 60 s, and 57 (87.7%) within 90 s.

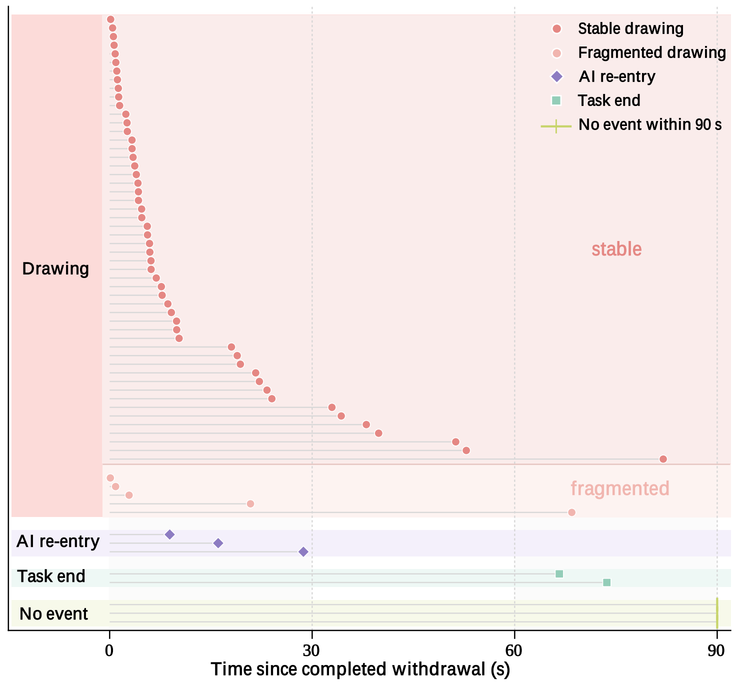

Fig. 4. Post-withdrawal activity following 65 completed withdrawals across 61 task records. Each row represents one completed withdrawal and is aligned to the end of the withdrawal transition ($t = 0$). The marker shows the earliest observed drawing action, child-initiated AI re-entry, or task end within the subsequent 90 s; crosses at 90 s indicate that none of these events occurred. Among the 57 withdrawals followed first by drawing, color distinguishes stable continuation ($n = 52$) from fragmented continuation ($n = 5$). Drawing observed only after AI re-entry was not counted as independent post-withdrawal continuation.

To distinguish sustained continuation from isolated or short-lived drawing activity, we further classified these 57 drawing-first episodes. For each episode i, stable continuation was defined as

$$\text{Stable}_i = \mathbf{1}\left[N_i \geq 3 \ \wedge \ \left(t_{i,N_i} - t_{i,1}\right) \geq 10\ \text{s}\right], \tag{1}$$

where N_i is the number of stroke or erasure initiations observed before re-entry, task end, or the end of the 90-second window, and $t_{i,j}$ is the time of the jth drawing action relative to withdrawal completion. Stable continuation therefore required at least three drawing actions spanning at least 10 s; episodes that did not meet both conditions were classified as fragmented. Requiring both conditions prevented an isolated mark or a brief cluster of operations from being interpreted as sustained continuation. Applying this definition, 52 of the 57 drawing-first episodes (91.2%) were stable and five (8.8%) were fragmented (Figure 4); stable continuation therefore followed 80.0% of all completed withdrawals.

The eight withdrawals not followed first by drawing comprised three child-initiated AI re-entries, two task endings, and three cases with no observed event within 90 s (Figure 4).

Overall, prompt and stable drawing predominated after completed withdrawal. In relation to RQ1, this extends the judgment-level account in Section 5.2.1: DrawBreath not only generated autonomous *Withdraw* judgments, but enacted withdrawals were usually followed by sustained child-led creation rather than immediate AI re-entry,

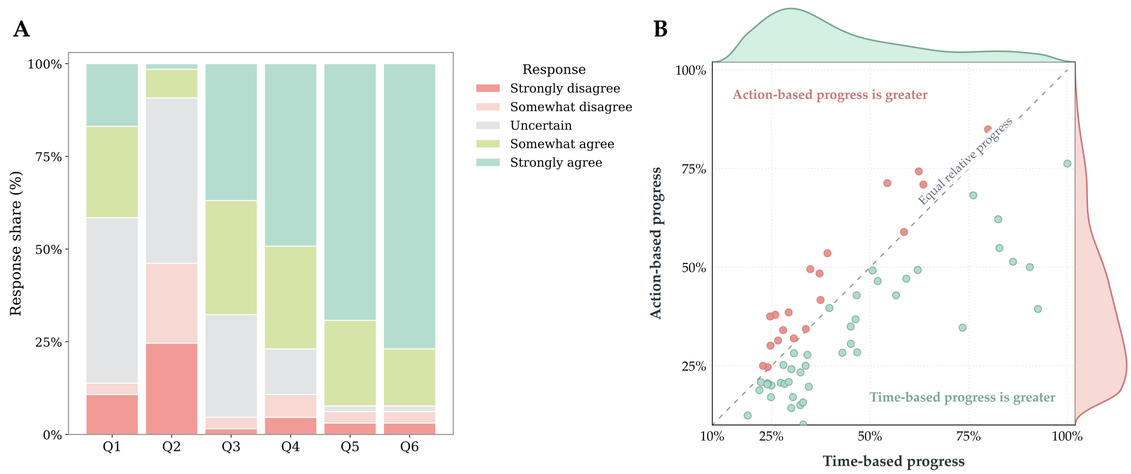

Fig. 5. Children's evaluations of AI withdrawal and the position of the first completed withdrawal. (A) Five-point response distributions for six post-task questionnaire items ($N = 65$; Table 3). Bars retain the original response directions. Q2 was negatively worded, so disagreement indicates a more positive evaluation for that item. (B) Time- and action-based positions of the first completed withdrawal across 61 task records. Each point represents one record; the top and right marginal density curves summarize the distributions along time- and action-based progress, respectively. The dashed diagonal denotes equal relative progress, with points above and below it indicating greater action-based and time-based progress. Both axes begin at 10%, reflecting the observed plotted range.

Table 3. The full questionnaire statements corresponding to Q1–Q6 in Figure 5A.

Item	**Questionnaire statement**
Q1	Most of the time, the AI assistant disappeared when I was already able to continue drawing on my own.
Q2	Sometimes, the AI assistant disappeared while I still needed to look at the advice it had just given me.
Q3	Overall, I felt that it was appropriate for the AI assistant to disappear automatically.
Q4	When the AI assistant disappeared, it did not noticeably interrupt the drawing I was working on.
Q5	After the AI assistant disappeared, I usually still knew what to draw next.
Q6	After the AI assistant disappeared, I was able to continue drawing based on my own ideas.

Note. Q2 was the only negatively worded item; disagreement with this statement therefore corresponds to a more positive evaluation of withdrawal timing.

task termination, or inactivity. These outcomes connect autonomous participation judgments to their observable consequence: foreground AI presence receded while creative responsibility remained with the child. They therefore provide evidence that withdrawal operated as a viable form of participation regulation, while not establishing a uniquely or optimally timed decision; post-withdrawal behavior cannot reveal whether an alternative timing would have been equally or more appropriate. We next examine children's post-task evaluations of withdrawal appropriateness and continuity.

5.2.3 Children's Perceptions of Withdrawal Appropriateness and Post-Withdrawal Continuity. Figure 5A presents the item-level response distributions for all 65 children, with full English translations of the questionnaire statements provided in Table 3. These task-level ratings complement the completed-withdrawal event analysis by showing how children evaluated the overall withdrawal experience.

Overall appropriateness was positively rated, whereas responses to the two items that most directly probed alignment with children's readiness to continue and ongoing need for support were less decisive (Figure 5A). For Q3, 44 of 65 children (67.7%) agreed or strongly agreed that automatic withdrawal was appropriate overall, while 3 (4.6%) disagreed or strongly disagreed. For Q1, 27 children (41.5%) agreed that AI usually withdrew once they could continue drawing on their own, 29 (44.6%) selected *Uncertain*, and 9 (13.8%) disagreed. For the negatively worded Q2, 30 children (46.2%) disagreed that the AI sometimes withdrew while they still needed its previous advice, 29 (44.6%) were uncertain, and 6 (9.2%) agreed. Because Q2 was negatively worded, disagreement represents a more positive assessment of withdrawal timing. The *Uncertain* category was the largest response group for both support-alignment items.

Children were considerably more decisive about perceived continuity after withdrawal (Figure 5A). Fifty of 65 children (76.9%) agreed that withdrawal did not noticeably interrupt their drawing. For Q5 and Q6, 60 children (92.3%) agreed that they still knew what to draw next and could continue with their own ideas, respectively; only one child (1.5%) selected *Uncertain* for each item. Strong agreement alone accounted for 45 responses (69.2%) to Q5 and 50 (76.9%) to Q6.

The questionnaire responses complement the post-withdrawal behavior reported in Section 5.2.2: the logs show that children typically returned promptly to the canvas and usually sustained stable drawing, while the ratings show that most children regarded withdrawal as appropriate overall, experienced little interruption, and felt able to continue. Children were nevertheless more decisive about post-withdrawal continuity than about whether withdrawal aligned with their support needs. In relation to RQ1, this convergence is consistent with the participation policy identifying moments at which it could reduce its foreground presence while leaving room for continued child-led work. It therefore supports treating restraint—regulating whether continued AI presence is still warranted—as a component of AI agency. This evidence does not identify a unique or precisely optimal withdrawal point: the behavioral logs and post-task ratings characterize what followed withdrawal and how it was experienced, rather than providing episode-level ground truth for individual withdrawal moments.

5.3 Understanding AI Withdrawal through the Lens of Children's Creative Trajectories

5.3.1 Locating AI Withdrawal within the Overall Creative Process. To begin the RQ2 analysis, we located the first completed withdrawal in each of the 61 task records with an enacted withdrawal. Using one anchor per record prevented the three records with repeated withdrawals from contributing more than once. Unlike the latency-from-availability analysis in Section 5.2.1, this analysis positioned withdrawal relative to each child's complete observed creative trajectory.

We characterized this position in two complementary ways:

$$P_i^{\text{time}} = \frac{t_i^W}{T_i}, \qquad P_i^{\text{op}} = \frac{O_i^W}{O_i^{\text{total}}}, \tag{2}$$

where t_i^W is the task-relative withdrawal-completion time and T_i is the actual task duration; O_i^W and O_i^{total} are the canvas-operation sequence values at withdrawal completion and task completion, respectively. P_i^{op} indexes accumulated drawing, revision, and tool-setting activity, not visual completeness. For three records without a snapshot at withdrawal completion, O_i^W was recovered from the timestamped operation log.

First completed withdrawals were concentrated toward the lower-left of Figure 5B, indicating that most occurred within the first half of both task time and action-based progress. The marginal distributions summarize this concentration separately along the two axes. The median temporal position was 34.1% of the actual task duration (IQR = 28.0–54.4%),

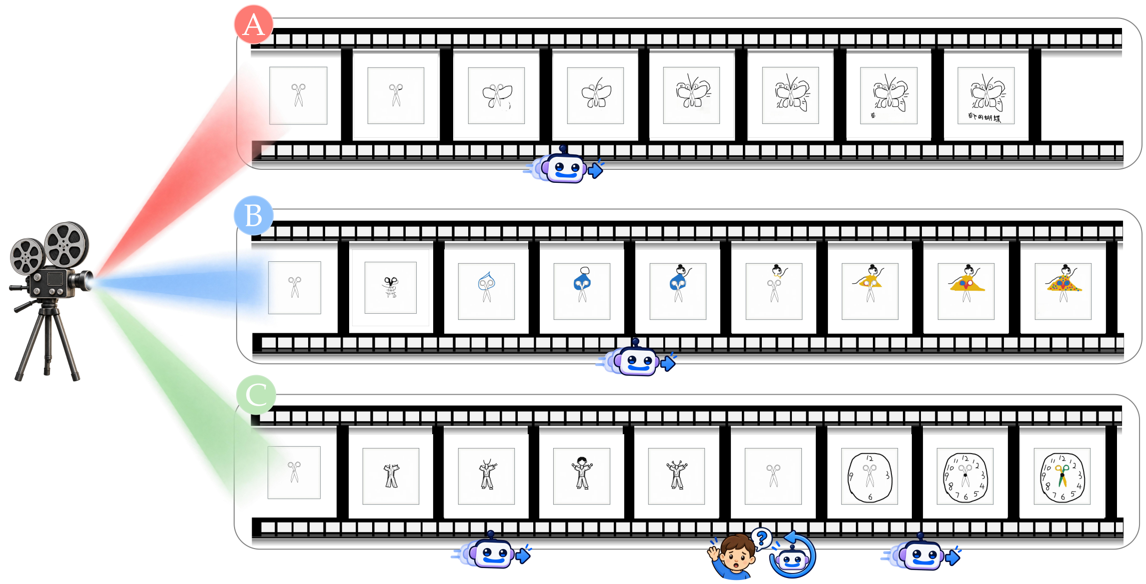

Fig. 6. Three contrasting visual trajectories surrounding AI withdrawal. Each row presents time-ordered canvas snapshots from left to right; icons mark completed AI withdrawal, a child help request, and AI re-entry. (A) Continued elaboration after visual-direction stabilization. (B) Revision within an established direction. (C) Later creative reorientation followed by renewed AI engagement and a subsequent withdrawal.

with 43 of 61 withdrawals (70.5%) occurring before the temporal midpoint. The median action-based position was 30.1% (IQR = 20.7–48.6%), with 47 withdrawals (77.0%) occurring before half of the final operation sequence had accumulated. Positions nevertheless extended across the plotted range and reached task completion on both measures, showing that withdrawal was not confined to a single stage.

The paired view further showed that the two measures were not interchangeable. Thirty-nine of the 61 tasks (63.9%) fell below the diagonal, where time-based progress exceeded action-based progress; 21 (34.4%) fell above it, and one (1.6%) lay on the diagonal. Across records, action-based progress trailed temporal progress by a median of 5.3 percentage points. Together, these findings provide a trajectory-level answer to RQ2: the first completed withdrawal was generally situated in the first half of children's creative processes, but its location varied across trajectories and according to whether progress was viewed through elapsed time or accumulated canvas activity. Withdrawal therefore did not occupy a fixed stage of creation. This overview establishes where it entered the broader creative process and provides the trajectory context for the finer-grained analyses that follow.

5.3.2 Visual Contexts Surrounding AI Withdrawal. Whereas the preceding analysis positioned withdrawal quantitatively within task time and canvas activity, the visual trajectories showed that withdrawal did not correspond to a single stage of drawing completion. Instead, the trajectories differed in both how stable the child's visual direction was at withdrawal and how that direction subsequently evolved. Figure 6 presents three contrasting trajectories: continued elaboration after a direction had stabilized (A), revision within an established direction (B), and later creative reorientation followed by renewed AI engagement (C).

In trajectory A, the child established a recognizable butterfly interpretation relatively early in the drawing process. By the time the AI withdrew, the core visual direction had stabilized, although the drawing itself remained incomplete. Subsequent snapshots added contours, internal details, and annotations without reconsidering what the drawing represented. This trajectory situates withdrawal at a transition from establishing a creative direction to child-led elaboration.

Trajectory B showed a different relationship between withdrawal and visual development. Although the child had already committed to a character-based interpretation, the visual realization of that idea remained under active revision. After withdrawal, previously drawn body and clothing elements were removed or substantially reworked while the underlying character concept persisted and was further elaborated. Visible revision therefore did not coincide with a loss of creative direction; here, withdrawal was situated within child-led experimentation around an established interpretation rather than only after visual stabilization.

Trajectory C further showed that a stable direction at withdrawal did not imply stability for the remainder of the creative process. At the initial withdrawal, the child had developed a recognizable human figure. The child later removed this interpretation, returned toward the starting form, and developed a substantially different clock-based composition. This reorientation was accompanied by a help request and AI re-entry, after which the AI subsequently withdrew again. The sequence reveals a boundary of treating withdrawal as terminal: independent progress at one moment did not preclude renewed support seeking later in the same creative process.

Taken together, these trajectories situate AI withdrawal across different relationships between visual-direction stability and subsequent creative change, rather than at a single notion of completion. Children could elaborate a stabilized direction, revise how an established idea was represented, or later abandon that direction and reorient the work. Withdrawal therefore marked a local transition in AI participation relative to the child's current creative state, not a terminal judgment that either the drawing or the child's need for support had permanently stabilized. This process-centered account complements the RQ1 analysis in Section 5.2. RQ1 examined whether DrawBreath could autonomously determine when to withdraw from foreground participation based on the unfolding creative process. RQ2 instead contextualizes the resulting withdrawal events, showing how they were embedded within visual trajectories that continued to elaborate, revise, or reorient after the AI stepped back.

6 Discussion

DrawBreath shows how an AI assistant can step back during children's creative work while remaining available when needed. Completed withdrawals were usually followed by sustained drawing, and most children reported that they could continue with their own ideas. The visual cases showed how withdrawal fitted into ongoing elaboration, revision, and changes of direction. Together, these findings suggest that AI participation can be adjusted throughout creation: help can remain accessible even when the assistant is no longer visible.

6.1 Understanding AI Withdrawal during Children's Creative Work

6.1.1 Stepping Back before and after Giving Help. Eleven of 65 children explicitly requested AI support, and 59 of 77 model-authored *Withdraw* judgments concerned unused presence. This pattern broadens the role of withdrawal beyond ending a support exchange. Scaffolding research describes fading as a gradual transfer of responsibility from the helper to the learner [44, 63]. In DrawBreath, withdrawal also occurred before help was requested. In these episodes, the system was deciding whether to keep the assistant visible while the child worked independently.

Mixed-initiative interaction considers the benefits, costs, and uncertainty of automated action [30, 37, 48], while COFI describes how initiative, contributions, and communication shape co-creation [35, 57]. Our design extends these questions to how long an assistant remains visible. This makes stepping back a part of participation design, alongside deciding when to offer a contribution or respond to a request.

The distinction also directs attention to how children seek help. Help-seeking depends on both the learner and the system [2, 58], and tutoring research examines how the helper's social presence and role relate to this process [31]. A child who has not asked a question may be making progress or may still be deciding what to ask. Withdrawal should therefore consider evidence that the child can proceed, alongside whether an exchange or request remains unresolved. This provides a practical basis for adjusting presence while keeping help within reach.

6.1.2 Withdrawing while Creation Continues. First completed withdrawals generally occurred before the midpoint of both task time and accumulated activity. The visual cases help interpret this early positioning: children continued elaborating a butterfly or revising a character's appearance while maintaining an established idea. Drawing research describes reciprocal influences between visual production and cognition [10, 21, 26, 64]; continued revision can thus develop an idea without necessarily indicating a need for external direction.

The child who replaced a human figure with a clock later sought support. This case illustrates how a child's need for help can change as the drawing develops. Withdrawal can leave room to pursue a current idea, with re-entry supporting a later question or change of direction. CoAuthor examines human–AI writing through sequences of requests, acceptance, dismissal, and revision [38]. Our cases extend this sequence perspective to the intervals when the assistant steps back and becomes available again. The resulting unit of interest is the unfolding creative process, including periods of independent work between exchanges.

This perspective complements access-timing research. Qin et al. found that early LLM access could reduce original ideas and perceived creative ownership in writing ideation [54]; Zhi et al. showed that timing effects on critical-thinking performance changed with time availability [72]. These studies connect the timing of assistance to the surrounding task. For children's drawing, our findings motivate examining how AI participation changes *after* access begins, as children develop, revise, or replace an idea.

6.1.3 Continuing to Draw and Knowing When to Step Back. Drawing preceded re-entry or task end after 57 of 65 completed withdrawals, with 52 meeting the stable-continuation criterion. At the task level, the statements about knowing what to draw next and continuing with their own ideas each received agreement from 60 of 65 children. These findings show that withdrawal was usually followed by continued child-led creation. Because the policy selected moments of independent progress, this pattern does not establish that withdrawal caused greater independence.

Responses about readiness to continue (Q1) and continued need for advice (Q2) were less decisive, with *Uncertain* the largest category. The contrast suggests that continuing an activity and judging the timing of assistance are distinct aspects of the experience. Advice use and the whole-task perspective of the questionnaire also matter when interpreting these responses. As Zhi et al. argue, different measures can reveal different aspects of collaboration [72]. For withdrawal, this distinction motivates combining records of continued work with children's accounts of when advice remained useful.

6.2 Design Implications for AI Participation in Children's Creative Activities

6.2.1 Keep Help within the Child's Reach. Withdrawal and re-entry should be designed together. Independent continuation and renewed help-seeking occurred within the same task, while three cancelled fades showed how children could interrupt withdrawal. Human–AI interaction guidelines emphasize user control and opportunities to correct system behavior [4, 35]. For withdrawal, this means that children retain the final say over when to return to the assistant. Coyle et al. show that assistance parameters can affect the experienced sense of agency [14]; the ease of restoring help and the child's understanding of that option therefore deserve attention alongside the availability of a control.

Children should be able to stop a fade, restore advice, or ask a new question without justifying their return. Returning to a previous exchange helps a child continue using an earlier suggestion; starting a new question supports a changed direction. Future interfaces could also let children keep selected advice visible or pause automatic withdrawal. These options would accommodate situations in which a child continues drawing while still referring to advice. Clear re-entry cues and few steps to restore help would make the child's control easier to exercise.

6.2.2 Wait for More Evidence When Support Needs Are Unclear. *Observe* comprised nearly half of model-authored judgments and immediately preceded the first *Withdraw* judgment in 16 of 20 deferred pathways. Because *Observe* and *Stay* shared the same interface, the assistant could remain available while the system waited for more evidence. This applies mixed-initiative reasoning under uncertainty [30] without adding advice or announcing every reassessment.

An ambiguous pause or revision can be a reason to observe what happens next. The system can retain access and reconsider withdrawal after an ongoing action or exchange. Notification research shows that attending to task boundaries can reduce disruption from system-initiated messages [1, 33], while proactive programming support highlights the importance of fitting assistance into an ongoing workflow [51]. Applied to withdrawal, these insights suggest that uncertainty can be handled through patience. Future comparisons could examine when this extra observation helps the assistant step back at a suitable moment.

6.2.3 Consider What the Child Can Do Next and the Time Available. The visual cases suggest that withdrawal can accompany both elaboration and substantial revision. A policy should consider whether the child has a direction to pursue, whether an exchange is still underway, and whether advice is being consulted. The assistance dilemma concerns when giving or withholding help benefits learning [6, 8, 34]. For creative work, this raises a related design question: when would stepping back give the child room to develop the next part of an idea?

Opportunity to seek help also matters. The median first *Withdraw* judgment occurred 29 seconds after access opened, following a two-minute independent opening and a 15-second availability safeguard. These timings place withdrawal within a task that already included independent work. Given that access-timing effects vary with available time [72], designers should consider the opening period, time to explore assistance, and remaining task time together. The same pause may have different implications near a deadline and during an extended drawing session.

The effort required to regain help is another consideration. In AI-assisted decision-making, Buçinca et al.'s cognitive forcing interventions reduced overreliance but received less favorable ratings, with benefits varying by cognitive motivation [12]. This trade-off motivates keeping withdrawal easy to reverse. Designs that encourage independent work should also make returning to support straightforward when the child chooses to do so.

6.3 Implications for Evaluating Human–AI Creative Collaboration

6.3.1 Evaluate Continuity, Reversibility, and Timing. The 77 *Withdraw* judgments, 68 initiated fades, and 65 completed withdrawals capture successive stages of the withdrawal process. Keeping these stages separate allows researchers to examine how model decisions, runtime safeguards, and children's actions jointly shape participation. Evaluation can then connect each completed transition to the child's subsequent activity and experience, while distinguishing an interface change from the child's acceptance of it.

Building on this distinction, we propose three complementary dimensions for evaluating withdrawal within an ongoing creative process.

Process continuity concerns whether children continue creating independently after the assistant steps back. Activity records can show when drawing resumes and how it develops before help is sought again. Following the sequence perspective of CoAuthor [38], analysis can include independent work between exchanges as well as direct interaction with the assistant.

Reversibility concerns whether children can restore help when they want it. Evaluation should examine whether children can find and use the re-entry option, the effort involved, and their experience of controlling the return. This connects the available control to experienced agency [14], making the usability of reversal an explicit evaluation question.

Timing appropriateness concerns whether withdrawal fits the child's support needs at that moment. The contrast between continuity and support-alignment ratings in our study motivates assessing these separately, consistent with work showing that different measures reveal different aspects of collaboration [72]. Brief retrospective discussion of selected transitions could clarify whether earlier advice was still useful or a later request arose from a new creative problem.

Together, these dimensions make changes in AI presence an explicit object of evaluation. They can complement established measures of creative experience [13, 25, 60] by explaining how children continue working, regain help, and experience the timing of participation changes.

6.3.2 Study Withdrawal and Re-entry as Part of AI Access. These dimensions also provide a basis for comparing participation policies. Access-timing research [54, 72] can be extended to compare continuous foreground presence, child-controlled dismissal, scheduled withdrawal, and process-contingent withdrawal. Keeping assistance capabilities and re-entry options consistent would help examine how each policy balances continued creation, access to help, and timing.

Vaccaro et al.'s meta-analysis distinguishes augmentation relative to humans alone from synergy relative to the better of humans or AI alone [62]. For child-led drawing, the comparison should similarly follow the research question: an AI-free condition examines what assistance adds, while a continuously visible assistant examines what withdrawal changes.

Comparisons should also distinguish unused-presence from post-support withdrawal, since reducing an unused assistant's visibility and stepping back after advice involve different support histories. The time available to seek help and the surrounding creative activity provide further context: equal fractions of task time did not correspond to equal accumulated action in our trajectories. Evaluating transitions in relation to both prior support and ongoing work would help identify when changes in AI presence support human control over assistance [28].

6.4 Limitations and Future Work

Population, duration, and implementation. The findings concern sixth-grade children at one public primary school completing a drawing task in a computer lab. Children's ages, familiarity with AI, and classroom practices may shape when they seek help and how they interpret an assistant stepping back [27, 40]. Extending the study to other age groups, schools, and creative media would help establish how withdrawal should adapt to these contexts.

Each session lasted up to 15 minutes. Longer creative projects can involve returning to an idea across sessions, consulting teachers or peers, and developing preferences through repeated use. Longitudinal classroom studies could examine how these experiences shape help-seeking and whether children come to prefer different levels of AI presence over time.

The findings also reflect a particular model, prompt, process summary, and sidebar interface. Other assistants may offer different advice or interpret the same activity differently, and creative tools may provide different ways to step back and return [66]. Replication across these configurations would clarify which aspects of withdrawal transfer across systems and which require adjustment to the task and interface.

7 Conclusion

This study shows that AI participation in children's creative work need not be defined by continuous presence. Through DrawBreath, we found that children usually continued creating after the AI stepped back, while *Withdraw* occurred both after explicit support and, more often, when available AI assistance had remained unused. Visual trajectories further showed that withdrawal was embedded within ongoing creative processes: children continued to elaborate and revise established ideas, sometimes reoriented their work, and could return to the AI when new support needs emerged. Our findings suggest that designing human–AI creative collaboration requires careful consideration not only of when AI should intervene, but also of when it should reduce its foreground presence while preserving an easy path back to support. We hope this work provides a foundation for broader investigations of adaptive AI participation and for designing creative AI systems that know not only when to contribute, but also when to do less.

References

[1] Piotr D. Adamczyk and Brian P. Bailey. 2004. If Not Now, When?: The Effects of Interruption at Different Moments within Task Execution. In *Proceedings of the SIGCHI Conference on Human Factors in Computing Systems (CHI '04)*. ACM, New York, NY, USA, 271–278. doi:10.1145/985692.985727

[2] Vincent Aleven, Elmar Stahl, Silke Schworm, Frank Fischer, and Raven Wallace. 2003. Help Seeking and Help Design in Interactive Learning Environments. *Review of Educational Research* 73, 3 (2003), 277–320. doi:10.3102/00346543073003277

[3] Teresa M. Amabile. 1982. Social Psychology of Creativity: A Consensual Assessment Technique. *Journal of Personality and Social Psychology* 43, 5 (1982), 997–1013. doi:10.1037/0022-3514.43.5.997

[4] Saleema Amershi, Dan Weld, Mihaela Vorvoreanu, Adam Fourney, Besmira Nushi, Penny Collisson, Jina Suh, Shamsi Iqbal, Paul N. Bennett, Kori Inkpen, Jaime Teevan, Ruth Kikin-Gil, and Eric Horvitz. 2019. Guidelines for Human–AI Interaction. In *Proceedings of the 2019 CHI Conference on Human Factors in Computing Systems (CHI '19)*. ACM, New York, NY, USA, 1–13. doi:10.1145/3290605.3300233

[5] Barrett R. Anderson, Jash Hemant Shah, and Max Kreminski. 2024. Homogenization Effects of Large Language Models on Human Creative Ideation. In *Proceedings of the 16th Conference on Creativity & Cognition (C&C '24)*. ACM, New York, NY, USA, 413–425. doi:10.1145/3635636.3656204

[6] Roger Azevedo, Jennifer G. Cromley, and Diane Seibert. 2004. Does Adaptive Scaffolding Facilitate Students' Ability to Regulate Their Learning with Hypermedia? *Contemporary Educational Psychology* 29, 3 (2004), 344–370. doi:10.1016/j.cedpsych.2003.09.002

[7] Baptiste Barbot. 2018. The Dynamics of Creative Ideation: Introducing a New Assessment Paradigm. *Frontiers in Psychology* 9 (2018), 2529. doi:10.3389/fpsyg.2018.02529

[8] Hamsa Bastani, Osbert Bastani, Alp Sungu, Haosen Ge, Özge Kabakcı, and Rei Mariman. 2025. Generative AI without Guardrails Can Harm Learning: Evidence from High School Mathematics. *Proceedings of the National Academy of Sciences* 122, 26 (2025), e2422633122. doi:10.1073/pnas.2422633122

[9] Benjamin Beltzung, Lison Martinet, Andrew J. J. MacIntosh, Xavier Meyer, Jérôme Hosselet, Marie Pelé, and Cédric Sueur. 2023. To Draw or Not to Draw: Understanding the Temporal Organization of Drawing Behavior Using Fractal Analyses. *Fractals* 31, 1 (2023), 2350009. doi:10.1142/S0218348X23500093

[10] Marion Botella, Vlad Petre Glăveanu, Franck Zenasni, Martin Storme, Nils Myszkowski, Marion Wolff, and Todd Lubart. 2013. How Artists Create: Creative Process and Multivariate Factors. *Learning and Individual Differences* 26 (2013), 161–170. doi:10.1016/j.lindif.2013.02.008

[11] Marion Botella, Franck Zenasni, and Todd Lubart. 2018. What Are the Stages of the Creative Process? What Visual Art Students Are Saying. *Frontiers in Psychology* 9 (2018), 2266. doi:10.3389/fpsyg.2018.02266

[12] Zana Buçinca, Maja Barbara Malaya, and Krzysztof Z. Gajos. 2021. To Trust or to Think: Cognitive Forcing Functions Can Reduce Overreliance on AI in AI-Assisted Decision-Making. *Proceedings of the ACM on Human-Computer Interaction* 5, CSCW1 (2021), 188:1–188:21. doi:10.1145/3449287

[13] Erin Cherry and Celine Latulipe. 2014. Quantifying the Creativity Support of Digital Tools through the Creativity Support Index. *ACM Transactions on Computer-Human Interaction* 21, 4 (2014), 21:1–21:25. doi:10.1145/2617588

[14] David Coyle, James Moore, Per Ola Kristensson, Paul Fletcher, and Alan Blackwell. 2012. I Did That! Measuring Users' Experience of Agency in Their Own Actions. In *Proceedings of the SIGCHI Conference on Human Factors in Computing Systems (CHI '12)*. ACM, New York, NY, USA, 2025–2034.

doi:10.1145/2207676.2208350

[15] Nicholas Davis, Chih-Pin Hsiao, Kunwar Yashraj Singh, Lisa Li, Sanat Moningi, and Brian Magerko. 2015. Drawing Apprentice: An Enactive Co-Creative Agent for Artistic Collaboration. In *Proceedings of the 2015 ACM SIGCHI Conference on Creativity and Cognition (C&C '15)*. ACM, New York, NY, USA, 185–186. doi:10.1145/2757226.2764555

[16] Nicholas Davis, Chih-Pin Hsiao, Kunwar Yashraj Singh, and Brian Magerko. 2016. Co-Creative Drawing Agent with Object Recognition. *Proceedings of the AAAI Conference on Artificial Intelligence and Interactive Digital Entertainment* 12, 1 (2016), 9–15. doi:10.1609/aiide.v12i1.12863

[17] Kees Dorst and Nigel Cross. 2001. Creativity in the Design Process: Co-Evolution of Problem–Solution. *Design Studies* 22, 5 (2001), 425–437. doi:10.1016/S0142-694X(01)00009-6

[18] Anil R. Doshi and Oliver P. Hauser. 2024. Generative AI Enhances Individual Creativity but Reduces the Collective Diversity of Novel Content. *Science Advances* 10, 28 (2024), eadn5290. doi:10.1126/sciadv.adn5290

[19] Fiona Draxler, Anna Werner, Florian Lehmann, Matthias Hoppe, Albrecht Schmidt, Daniel Buschek, and Robin Welsch. 2024. The AI Ghostwriter Effect: When Users Do Not Perceive Ownership of AI-Generated Text but Self-Declare as Authors. *ACM Transactions on Computer-Human Interaction* 31, 2 (2024), 25:1–25:40. doi:10.1145/3637875

[20] Stefania Druga and Amy J. Ko. 2025. Scratch Copilot: Supporting Youth Creative Coding with AI. In *Proceedings of the 24th Interaction Design and Children Conference (IDC '25)*. ACM, New York, NY, USA, 140–153. doi:10.1145/3713043.3727051

[21] Judith E. Fan, Wilma A. Bainbridge, Rebecca Chamberlain, and Jeffrey D. Wammes. 2023. Drawing as a Versatile Cognitive Tool. *Nature Reviews Psychology* 2, 9 (2023), 556–568. doi:10.1038/s44159-023-00212-w

[22] Judith E. Fan, Monica Dinculescu, and David Ha. 2019. CollabDraw: An Environment for Collaborative Sketching with an Artificial Agent. In *Proceedings of the 2019 Conference on Creativity and Cognition (C&C '19)*. ACM, New York, NY, USA, 556–561. doi:10.1145/3325480.3326578

[23] Min Fan, Xinyue Cui, Jing Hao, Renxuan Ye, Wanqing Ma, Xin Tong, and Meng Li. 2024. StoryPrompt: Exploring the Design Space of an AI-Empowered Creative Storytelling System for Elementary Children. In *Extended Abstracts of the 2024 CHI Conference on Human Factors in Computing Systems (CHI EA '24)*. ACM, New York, NY, USA, 1–8. doi:10.1145/3613905.3651118

[24] Min Fan, Xinyue Cui, Wanqing Ma, Haiyan Li, Xin Tong, Lin Yang, and Yonghui Wang. 2025. From Words to Wonder: Designing and Evaluating an AI-Empowered Creative Storytelling System for Elementary Children. In *Proceedings of the 2025 CHI Conference on Human Factors in Computing Systems (CHI '25)*. ACM, New York, NY, USA, 1–15. doi:10.1145/3706598.3713478

[25] Jonas Frich, Lindsay MacDonald Vermeulen, Christian Remy, Michael Mose Biskjaer, and Peter Dalsgaard. 2019. Mapping the Landscape of Creativity Support Tools in HCI. In *Proceedings of the 2019 CHI Conference on Human Factors in Computing Systems (CHI '19)*. ACM, New York, NY, USA, 1–18. doi:10.1145/3290605.3300619

[26] Gabriela Goldschmidt. 1991. The Dialectics of Sketching. *Creativity Research Journal* 4, 2 (1991), 123–143. doi:10.1080/10400419109534381

[27] Ariel Han, Xiaofei Zhou, Zhenyao Cai, Shenshen Han, Richard Ko, Seth Corrigan, and Kylie A. Peppler. 2024. Teachers, Parents, and Students' Perspectives on Integrating Generative AI into Elementary Literacy Education. In *Proceedings of the 2024 CHI Conference on Human Factors in Computing Systems (CHI '24)*. ACM, New York, NY, USA, 1–17. doi:10.1145/3613904.3642438

[28] Jeffrey Heer. 2019. Agency Plus Automation: Designing Artificial Intelligence into Interactive Systems. *Proceedings of the National Academy of Sciences* 116, 6 (2019), 1844–1850. doi:10.1073/pnas.1807184115

[29] Jimpei Hitsuwari, Yoshiyuki Ueda, Woojin Yun, and Michio Nomura. 2023. Does Human–AI Collaboration Lead to More Creative Art? Aesthetic Evaluation of Human-Made and AI-Generated Haiku Poetry. *Computers in Human Behavior* 139 (2023), 107502. doi:10.1016/j.chb.2022.107502

[30] Eric Horvitz. 1999. Principles of Mixed-Initiative User Interfaces. In *Proceedings of the SIGCHI Conference on Human Factors in Computing Systems (CHI '99)*. ACM, New York, NY, USA, 159–166. doi:10.1145/302979.303030

[31] Iris Howley, Takayuki Kanda, Kotaro Hayashi, and Carolyn Rosé. 2014. Effects of Social Presence and Social Role on Help-Seeking and Learning. In *Proceedings of the 2014 ACM/IEEE International Conference on Human-Robot Interaction (HRI '14)*. ACM, New York, NY, USA, 415–422. doi:10.1145/2559636.2559667

[32] Zhejing Hu, Yan Liu, Zhi Zhang, Gong Chen, Bruce X. B. Yu, Junxian Li, and Jiannong Cao. 2026. MusicScaffold: Bridging Machine Efficiency and Human Growth in Adolescent Creative Education through Generative AI. In *Proceedings of the 2026 CHI Conference on Human Factors in Computing Systems (CHI '26)*. ACM, New York, NY, USA, 1–21. doi:10.1145/3772318.3790440

[33] Shamsi T. Iqbal and Brian P. Bailey. 2008. Effects of Intelligent Notification Management on Users and Their Tasks. In *Proceedings of the SIGCHI Conference on Human Factors in Computing Systems (CHI '08)*. ACM, New York, NY, USA, 93–102. doi:10.1145/1357054.1357070

[34] Kenneth R. Koedinger and Vincent Aleven. 2007. Exploring the Assistance Dilemma in Experiments with Cognitive Tutors. *Educational Psychology Review* 19, 3 (2007), 239–264. doi:10.1007/s10648-007-9049-0

[35] Tomas Lawton, Francisco J. Ibarrola, Dan Ventura, and Kazjon Grace. 2023. Drawing with Reframer: Emergence and Control in Co-Creative AI. In *Proceedings of the 28th International Conference on Intelligent User Interfaces (IUI '23)*. ACM, New York, NY, USA, 264–277. doi:10.1145/3581641.3584095

[36] Zoë Lazar-Kurz and Steven Brown. 2025. Trajectories of the Creative Process: A Multiple-Drafts Analysis of Graphic Design. *Creativity Research Journal* 37, 4 (2025), 736–754. doi:10.1080/10400419.2024.2350843

[37] John D. Lee and Katrina A. See. 2004. Trust in Automation: Designing for Appropriate Reliance. *Human Factors* 46, 1 (2004), 50–80. doi:10.1518/hfes.46.1.50_30392

[38] Mina Lee, Percy Liang, and Qian Yang. 2022. CoAuthor: Designing a Human–AI Collaborative Writing Dataset for Exploring Language Model Capabilities. In *Proceedings of the 2022 CHI Conference on Human Factors in Computing Systems (CHI '22)*. ACM, New York, NY, USA, 1–19.

doi:10.1145/3491102.3502030

[39] Yiren Liu, Si Chen, Haocong Cheng, Mengxia Yu, Xiao Ran, Andrew Mo, Yiliu Tang, and Yun Huang. 2024. How AI Processing Delays Foster Creativity: Exploring Research Question Co-Creation with an LLM-Based Agent. In *Proceedings of the 2024 CHI Conference on Human Factors in Computing Systems (CHI '24)*. ACM, New York, NY, USA, 1–25. doi:10.1145/3613904.3642698

[40] Duri Long and Brian Magerko. 2020. What Is AI Literacy? Competencies and Design Considerations. In *Proceedings of the 2020 CHI Conference on Human Factors in Computing Systems (CHI '20)*. ACM, New York, NY, USA, 1–16. doi:10.1145/3313831.3376727

[41] Todd I. Lubart. 2001. Models of the Creative Process: Past, Present and Future. *Creativity Research Journal* 13, 3–4 (2001), 295–308. doi:10.1207/S15326934CRJ1334_07

[42] Mary-Anne Mace and Tony Ward. 2002. Modeling the Creative Process: A Grounded Theory Analysis of Creativity in the Domain of Art Making. *Creativity Research Journal* 14, 2 (2002), 179–192. doi:10.1207/S15326934CRJ1402_5

[43] Xiaoyang Mao, Omar Galil, Quintcey Parrish, and Chiradeep Sen. 2020. Evidence of Cognitive Chunking in Freehand Sketching during Design Ideation. *Design Studies* 67 (2020), 1–26. doi:10.1016/j.destud.2019.11.009

[44] Katherine L. McNeill, David J. Lizotte, Joseph Krajcik, and Ronald W. Marx. 2006. Supporting Students' Construction of Scientific Explanations by Fading Scaffolds in Instructional Materials. *The Journal of the Learning Sciences* 15, 2 (2006), 153–191. doi:10.1207/s15327809jls1502_1

[45] Michele Newman, Kaiwen Sun, Ilena B. Dalla Gasperina, Grace Y. Shin, Matthew Kyle Pedraja, Ritesh Kanchi, Maia B. Song, Rannie Li, Jin Ha Lee, and Jason Yip. 2024. I Want It to Talk Like Darth Vader: Helping Children Construct Creative Self-Efficacy with Generative AI. In *Proceedings of the 2024 CHI Conference on Human Factors in Computing Systems (CHI '24)*. ACM, New York, NY, USA, 1–18. doi:10.1145/3613904.3642492

[46] Changhoon Oh, Jungwoo Song, Jinhan Choi, Seonghyeon Kim, Sungwoo Lee, and Bongwon Suh. 2018. I Lead, You Help but Only with Enough Details: Understanding User Experience of Co-Creation with Artificial Intelligence. In *Proceedings of the 2018 CHI Conference on Human Factors in Computing Systems (CHI '18)*. ACM, New York, NY, USA, 1–13. doi:10.1145/3173574.3174223

[47] Adiba Orzikulova, Han Xiao, Zhipeng Li, Yukang Yan, Yuntao Wang, Yuanchun Shi, Marzyeh Ghassemi, Sung-Ju Lee, Anind K. Dey, and Xuhai Xu. 2024. Time2Stop: Adaptive and Explainable Human–AI Loop for Smartphone Overuse Intervention. In *Proceedings of the 2024 CHI Conference on Human Factors in Computing Systems (CHI '24)*. ACM, New York, NY, USA, 1–20. doi:10.1145/3613904.3642747

[48] Raja Parasuraman, Thomas B. Sheridan, and Christopher D. Wickens. 2000. A Model for Types and Levels of Human Interaction with Automation. *IEEE Transactions on Systems, Man, and Cybernetics–Part A: Systems and Humans* 30, 3 (2000), 286–297. doi:10.1109/3468.844354

[49] John D. Patterson, Baptiste Barbot, James Lloyd-Cox, and Roger E. Beaty. 2024. AuDrA: An Automated Drawing Assessment Platform for Evaluating Creativity. *Behavior Research Methods* 56, 4 (2024), 3619–3636. doi:10.3758/s13428-023-02258-3

[50] Roy D. Pea. 2004. The Social and Technological Dimensions of Scaffolding and Related Theoretical Concepts for Learning, Education, and Human Activity. *The Journal of the Learning Sciences* 13, 3 (2004), 423–451. doi:10.1207/s15327809jls1303_6

[51] Kevin Pu, Daniel Lazaro, Ian Arawjo, Haijun Xia, Ziang Xiao, Tovi Grossman, and Yan Chen. 2025. Assistance or Disruption? Exploring and Evaluating the Design and Trade-Offs of Proactive AI Programming Support. In *Proceedings of the 2025 CHI Conference on Human Factors in Computing Systems (CHI '25)*. ACM, New York, NY, USA, 1–21. doi:10.1145/3706598.3713357

[52] Kevin Pu, Ting Zhang, Naveen Sendhilnathan, Sebastian Freitag, Raj Sodhi, and Tanya R. Jonker. 2025. ProMemAssist: Exploring Timely Proactive Assistance through Working Memory Modeling in Multi-Modal Wearable Devices. In *Proceedings of the 38th Annual ACM Symposium on User Interface Software and Technology (UIST '25)*. ACM, New York, NY, USA, 1–19. doi:10.1145/3746059.3747770

[53] Sadhana Puntambekar and Roland Hübscher. 2005. Tools for Scaffolding Students in a Complex Learning Environment: What Have We Gained and What Have We Missed? *Educational Psychologist* 40, 1 (2005), 1–12. doi:10.1207/s15326985ep4001_1

[54] Peinuan Qin, Chi-Lan Yang, Jingshu Li, Jing Wen, and Yi-Chieh Lee. 2025. Timing Matters: How Using LLMs at Different Timings Influences Writers' Perceptions and Ideation Outcomes in AI-Assisted Ideation. In *Proceedings of the 2025 CHI Conference on Human Factors in Computing Systems (CHI '25)*. ACM, New York, NY, USA, 1–16. doi:10.1145/3706598.3713146

[55] Brian J. Reiser. 2004. Scaffolding Complex Learning: The Mechanisms of Structuring and Problematizing Student Work. *The Journal of the Learning Sciences* 13, 3 (2004), 273–304. doi:10.1207/s15327809jls1303_2

[56] Mitchel Resnick. 2007. All I Really Need to Know (About Creative Thinking) I Learned (By Studying How Children Learn) in Kindergarten. In *Proceedings of the 6th ACM SIGCHI Conference on Creativity & Cognition (C&C '07)*. ACM, New York, NY, USA, 1–6. doi:10.1145/1254960.1254961

[57] Jeba Rezwana and Mary Lou Maher. 2023. Designing Creative AI Partners with COFI: A Framework for Modeling Interaction in Human–AI Co-Creative Systems. *ACM Transactions on Computer-Human Interaction* 30, 5 (2023), 67:1–67:28. doi:10.1145/3519026

[58] Ido Roll, Vincent Aleven, Bruce M. McLaren, and Kenneth R. Koedinger. 2011. Improving Students' Help-Seeking Skills Using Metacognitive Feedback in an Intelligent Tutoring System. *Learning and Instruction* 21, 2 (2011), 267–280. doi:10.1016/j.learninstruc.2010.07.004

[59] Mark A. Runco and Garrett J. Jaeger. 2012. The Standard Definition of Creativity. *Creativity Research Journal* 24, 1 (2012), 92–96. doi:10.1080/10400419.2012.650092

[60] Ben Shneiderman. 2007. Creativity Support Tools: Accelerating Discovery and Innovation. *Commun. ACM* 50, 12 (2007), 20–32. doi:10.1145/1323688.1323689

[61] Gerben Tolkamp, Bart Verwaeren, Tim Vriend, Aart-Jan Riekhoff, and Bernard Nijstad. 2025. Creativity as It Unfolds: An Examination of Temporality in the Creative Process. *Creativity Research Journal* 37, 1 (2025), 22–43. doi:10.1080/10400419.2023.2234719

[62] Michelle Vaccaro, Abdullah Almaatouq, and Thomas Malone. 2024. When Combinations of Humans and AI Are Useful: A Systematic Review and Meta-Analysis. *Nature Human Behaviour* 8, 12 (2024), 2293–2303. doi:10.1038/s41562-024-02024-1

[63] Janneke van de Pol, Monique Volman, and Jos Beishuizen. 2010. Scaffolding in Teacher–Student Interaction: A Decade of Research. *Educational Psychology Review* 22, 3 (2010), 271–296. doi:10.1007/s10648-010-9127-6

[64] Ilse M. Verstijnen, Cees van Leeuwen, Gabriela Goldschmidt, Ronald Hamel, and John M. Hennessey. 1998. Sketching and Creative Discovery. *Design Studies* 19, 4 (1998), 519–546. doi:10.1016/S0142-694X(98)00017-9

[65] David Wood, Jerome S. Bruner, and Gail Ross. 1976. The Role of Tutoring in Problem Solving. *Journal of Child Psychology and Psychiatry* 17, 2 (1976), 89–100. doi:10.1111/j.1469-7610.1976.tb00381.x

[66] Qian Yang, Aaron Steinfeld, Carolyn Rosé, and John Zimmerman. 2020. Re-examining Whether, Why, and How Human–AI Interaction Is Uniquely Difficult to Design. In *Proceedings of the 2020 CHI Conference on Human Factors in Computing Systems (CHI '20)*. ACM, New York, NY, USA, 1–13. doi:10.1145/3313831.3376301

[67] Yiwen Yin, Mingze Wu, Ruijie Huang, Xin Tong, Junyu Zhou, Chun Yu, and Yuanchun Shi. 2026. Proactive AI as a Catalyst for Creativity? Balancing Human Agency and AI Contribution in Collaborative Story Writing. In *Proceedings of the 2026 CHI Conference on Human Factors in Computing Systems (CHI '26)*. ACM, New York, NY, USA, 1–16. doi:10.1145/3772318.3790848

[68] Shun Yokochi and Takeshi Okada. 2005. Creative Cognitive Process of Art Making: A Field Study of a Traditional Chinese Ink Painter. *Creativity Research Journal* 17, 2–3 (2005), 241–255. doi:10.1080/10400419.2005.9651482

[69] Ann Yuan, Andy Coenen, Emily Reif, and Daphne Ippolito. 2022. Wordcraft: Story Writing With Large Language Models. In *Proceedings of the 27th International Conference on Intelligent User Interfaces (IUI '22)*. ACM, New York, NY, USA, 841–852. doi:10.1145/3490099.3511105

[70] Chao Zhang, Xuechen Liu, Katherine Ziska, Soobin Jeon, Chi-Lin Yu, and Ying Xu. 2024. Mathemyths: Leveraging Large Language Models to Teach Mathematical Language through Child–AI Co-Creative Storytelling. In *Proceedings of the 2024 CHI Conference on Human Factors in Computing Systems (CHI '24)*. ACM, New York, NY, USA, 1–23. doi:10.1145/3613904.3642647

[71] Chao Zhang, Cheng Yao, Jiayi Wu, Weijia Lin, Lijuan Liu, Ge Yan, and Fangtian Ying. 2022. StoryDrawer: A Child–AI Collaborative Drawing System to Support Children's Creative Visual Storytelling. In *Proceedings of the 2022 CHI Conference on Human Factors in Computing Systems (CHI '22)*. ACM, New York, NY, USA, 1–15. doi:10.1145/3491102.3501914

[72] Jiayin Zhi, Harsh Kumar, and Mina Lee. 2026. Investigating the Effects of LLM Use on Critical Thinking Under Time Constraints: Access Timing and Time Availability. In *Proceedings of the 2026 CHI Conference on Human Factors in Computing Systems (CHI '26)*. ACM, New York, NY, USA, 1–21. doi:10.1145/3772318.3791796

[73] Eric Zhou and Dokyun Lee. 2024. Generative Artificial Intelligence, Human Creativity, and Art. *PNAS Nexus* 3, 3 (2024), pgae052. doi:10.1093/pnasnexus/pgae052

A Scope and Reporting Conventions

These supplementary materials extend the main manuscript with implementation and audit details. Appendix B documents the formal-study configuration and participant flow; Appendix C reports the deployment architecture, retained server and hardware configuration, and frozen model settings; Appendix D gives the decision prompt and output contract; Appendix E gives the student-facing support prompt and per-turn payload; and Appendix F records the analysis units and reference denominators.

Terminology follows the paper's three-layer architecture: a *model-authored judgment* is an output of the Decision Agent, a *runtime assignment* is a system control record, and a *completed withdrawal* is an enacted interface transition. These units are not interchangeable.

B Formal Study Configuration

B.1 Participant Flow

We recruited 72 sixth-grade students from a public primary school. Seven were excluded because they did not complete the study session or submitted a blank response, leaving 65 participants and 65 analyzed task records. The sample was approximately balanced by gender. Participation was voluntary, the study protocol received approval from the authors' institution, and legal-guardian consent and child assent were obtained before participation. Students were recruited from two intact sixth-grade classes through a partner teacher. Interested families received an information sheet. Prior experience with conversational AI was not an eligibility requirement; students needed only to be able to use the on-screen drawing tools and enter a basic question. Sessions took place during regular art or information-technology periods, with each child using an individual account. Children received a small school-supplied gift and could stop at any time without an effect on their course grade.

Table 4. Participant flow for the formal study.

Stage	Count	Definition
Recruited	72	Sixth-grade students invited into the formal study.
Excluded	7	Did not complete the session or provided a blank submission; the available records do not support a more detailed numerical split.
Analyzed	65	One valid formal-study task record per child.

The de-identified system export does not contain age or gender fields. Accordingly, we do not report exact age statistics or exact gender counts in this appendix. This avoids introducing participant characteristics that cannot be verified against the supplied formal-study data package.

B.2 Task and Procedure

Each child completed one Open-ended Creative Drawing Continuation task on a desktop workstation in the school's computer lab. A single abstract or representational starting shape was randomly assigned and displayed on an otherwise blank canvas. The child was asked to incorporate, extend, or reinterpret that shape in an original drawing; no target image or uniquely correct completion was specified.

The task lasted up to 15 minutes. The first two minutes counted toward this limit and formed an AI-free opening period. When AI access opened, the runtime maintained an *Observe* hold for a 15-second minimum-availability period before a model-authored withdrawal could be considered. Children could submit before the deadline; otherwise, the system submitted the current drawing when time elapsed. The post-task questionnaire was administered once after submission.

Table 5. Fixed task and interface timings used in the formal study.

Setting	Value	Role
Nominal task limit	900 s	Maximum task duration
AI-free opening	120 s	Included in the 900-s limit
Minimum availability after access opened	15 s	Runtime safeguard; withdrawal ineligible
Hover-entry delay	450 ms	Low-salience message reveal
Pointer-leave delay	650 ms	Prevented abrupt remasking
Cancellable withdrawal fade	2.5 s	Interface transition
Post-withdrawal observation window	90 s	Offline analysis only

C System and Model Configuration

C.1 Deployment Architecture

DrawBreath used a React interface built with Vite, an Express service running on Node.js 20, and PostgreSQL 16. The drawing workspace and toolbar remained available throughout the task; adaptive participation affected only the AI sidebar. The service stored task records, time-stamped canvas operations, snapshots, assistant messages and events, support cycles, model requests and responses, validation outcomes, and withdrawal lifecycle events. Model inference was delegated to a hosted OpenAI-compatible Chat Completions endpoint, so the study server performed no local model inference. The application was containerized with Docker and Docker Compose. The deployment comprised an application container for the API and static front-end and a database container, each with a health check. The application listened on port 3300 and was exposed under a configured base-path prefix behind the reverse proxy. Source code is available at https://draw-breath.github.io/.

During the study, DrawBreath ran on the same dedicated server as the companion ideation subsystem. The following values were read from the study deployment host and describe that deployment rather than minimum system requirements:

- **Compute:** 4 virtual CPU cores and 8 GB RAM (`4c8g`; 7.3 GiB usable), 180 GB block storage, and a 12 Mbps public-network bandwidth tier (`12m`).
- **Operating system:** Ubuntu 22.04 LTS (kernel 5.15.0-88-generic, x86_64).
- **Container runtime:** Docker 25.0.2 with Docker Compose v2.24.5.
- **Application runtimes:** Node.js 20 (`bookworm-slim`) and PostgreSQL 16 (`alpine`) ran inside their respective containers rather than being installed directly on the host.
- **Client access:** the teacher console and student workspaces connected to the deployment over HTTPS.

C.2 Frozen Participation-Policy Settings

Table 6 separates online configuration from the 90-second post-withdrawal window used only in offline analysis. Recent-window metrics and reference signals summarized the unfolding process and determined when evidence was available for review. They did not mechanically assign a *Stay*, *Observe*, or *Withdraw* judgment.

Table 6. Participation-policy values retained in the formal-session configuration snapshots.

Field	**Value**	**Interpretation**
`aiPolicy`	`adaptive`	Participation policy
`ruleVersion`	`withdrawal_rules_v1`	Frozen configuration version
`flowWindowSeconds`	30 s	Canvas-evidence window
`flowEvaluationIntervalSeconds`	15 s	Reference-signal update interval
`flowIdleSeconds`	10 s	Reference-signal parameter
`aiAccessQuietSeconds`	20 s	Access-context parameter
`withdrawalCandidateSeconds`	15 s	Candidate-context parameter
`flowConsecutivePasses`	3	Reference-signal history length
`aiReopenCooldownSeconds`	180 s	Re-entry context
`snapshotIntervalSeconds`	30 s	Periodic snapshot interval

The Decision Agent received a review opportunity approximately every 30 seconds or after a meaningful interaction event. Evidence summaries covered approximately 8–15 seconds of immediate activity and 20–30 seconds of short-term development, with 60 seconds retained as broader context. A pause, erase action, undo, or reference-signal failure was never sufficient on its own to authorize withdrawal.

C.3 Model and Decision-Agent Decoding

Both the student-facing support component and the Decision Agent used GPT-5.5. The 267 non-fallback model-authored judgments in the analyzed data all retained `gpt-5.5` as the model identifier. Decision calls used the settings in Table 7; these values match the frozen formal-study description in the manuscript.

Table 7. Decision-Agent request and decoding settings.

Setting	Value
Endpoint	OpenAI-compatible Chat Completions API
Model	`gpt-5.5`
Temperature	0.2
Top-p	1.0
Frequency / presence penalty	0 / 0
Fixed seed	None
Explicit output-token cap	None
Response format	Structured JSON
Request timeout	30 s

D Decision Prompt and Output Contract

D.1 System Prompt

The following text is the system prompt retained with the formal-study decision calls:

Decision-Agent system prompt

You are the AI scaffolding-timing decision maker in a children's open-ended drawing-continuation task. You only determine whether the current action should be STAY, OBSERVE, WITHDRAW, or CANCEL_WITHDRAWAL; you do not generate content for the student. Canvas, conversation, access, and timing data are interpretable reference evidence, not hard rules. Do not withdraw while the student is actively typing, reading, or waiting for a response. When there is no direct support need and credible independent progress is present, prefer making room promptly. Never interpret a single undo, erasure, or brief pause as evidence of the student's ability or psychological state. Return strict JSON only.

The fourth protocol value, `CANCEL_WITHDRAWAL`, was a transition-control action for an active fade rather than a stable participation judgment. The analysis therefore reports the three model-authored participation judgments *Stay*, *Observe*, and *Withdraw*. No model-authored `CANCEL_WITHDRAWAL` occurred among the 267 analyzed judgments.

D.2 Evidence Packet and Return Schema

Each user message contained a JSON evidence packet. Its main groups were task time and configuration version; assistant visibility, quiet time, pending responses, panel activity, and input activity; support-cycle status; recent canvas metrics; reference signals; recent conversation; and data-quality information. When available, the request also carried the current canvas snapshot. The reference signals were contextual summaries, not labels or hard withdrawal rules.

The response contract was:

```
{
 "action":"STAY|OBSERVE|WITHDRAW|CANCEL_WITHDRAWAL",
```

```
 "confidence":0.0,
 "decisionSummary":"no more than 160 characters",
 "supportNeed":"LOW|MEDIUM|HIGH|UNCERTAIN",
 "supportEpisodeStatus":
   "NOT_USED|UNRESOLVED|PARTIALLY_RESOLVED|RESOLVED",
 "withdrawalType":"NONE|UNUSED_PRESENCE|POST_SUPPORT",
 "evidenceForWithdrawal":[],
 "evidenceAgainstWithdrawal":[],
 "uncertainties":[],
 "nextReviewSeconds":3,
 "studentVisibleMessage":null
}
```

A proposed `WITHDRAW` passed response validation only when `evidenceForWithdrawal` was non-empty and one of two pathways was internally consistent:

(1) **Unused-presence withdrawal:** `supportEpisodeStatus=NOT_USED` and `withdrawalType=UNUSED_PRESENCE`.
(2) **Post-support withdrawal:** `supportEpisodeStatus=PARTIALLY_RESOLVED` or `RESOLVED`, and `withdrawalType=POST_SUPPORT`.

An invalid withdrawal proposal was normalized to `OBSERVE`. A valid proposal could still remain unexecuted if the runtime was already handling a withdrawal or if another interaction safeguard applied. Thus, model output, validation, fade initiation, fade cancellation, and completed withdrawal were logged as separate events.

E Student-Facing Drawing Support

E.1 System Prompt Translation

The student-facing component answered only after a child submitted a predefined or free-text question. The following English translation preserves the complete instruction content of the system prompt used in the formal study; headings and lists are typeset for readability.

Student-facing support system prompt (English translation)

You are a Creative Drawing Continuation Learning Assistant for students. In an open-ended drawing task based on one or more starter shapes, provide help that is appropriately limited, restrained, and thought-provoking.

Task background

The student sees one or more existing abstract lines, geometric figures, incomplete shapes, or outlines of concrete objects. The student must preserve and use those starter shapes, continue drawing from them, and form a complete work with a personal idea and a coherent theme. There is no single correct answer. What matters is the student's idea and distinctive use of the starter shapes, not technical drawing skill, precise lines, or how polished the picture looks.

Your role

You are learning scaffolding and a thinking partner. You are not a drawing tool, answer generator, or artwork grader. Your tasks are to:

(1) help the student observe the existing shapes;

(2) help the student express an initial idea clearly;
(3) offer limited inspiration when the student is stuck;
(4) help compare different directions;
(5) help discover possible relationships among visual elements; and
(6) stop helping promptly when the student knows what to draw next, allowing independent work to resume.

Basic interaction rules

(1) Reply only when the student actively asks a question; do not proactively push ideas, compositions, or evaluations.
(2) Prefer one short question that helps the student think before offering a hint when needed.
(3) Focus each reply on the student's current single question; do not solve the whole drawing at once.
(4) Keep replies to 2–4 sentences whenever possible, using language students can understand.
(5) Do not generate a complete picture plan, complete story, detailed composition steps, or final artwork description.
(6) Do not directly tell the student what the drawing should become.
(7) Do not make the final choice for the student; leave the decision to them.
(8) Do not invoke a standard answer, best solution, or correct way to draw.
(9) Do not judge whether the student draws beautifully, professionally, or skillfully.
(10) Do not infer weak creativity merely because there are few lines, simple content, or temporarily low completion.
(11) Do not encourage simply adding details, lines, or filling the page as a way to increase creativity.
(12) Do not reveal automatic scores, creativity scores, grades, or rankings.
(13) Do not treat automatic scoring as a final conclusion about the student's creativity.
(14) Do not use discouraging, comparative, or labeling language such as "your creativity is low" or "other people usually draw more richly."
(15) Answer only questions related to the current idea, drawing, canvas operation, and creative expression. For an unrelated question, briefly say it is not closely related to the current drawing task and return the conversation to the artwork.

Scaffolding sequence

Step 1: identify the type of problem. The student may not know what the lines resemble; may have an idea but not know how to continue; may not know how several shapes connect; may be unsure about subject and background; may worry the idea is ordinary; may not know how to make the picture complete; may have a local drawing or software problem; may want to continue earlier advice; or may have entered a new stage with a new question.

Step 2: ask about the student's existing idea first, with no more than one central question at a time.

Step 3: provide only the minimum necessary hint, selecting at most one relevant private scaffolding dimension:

- a different way to see the starter shape;
- what part of an object it could become;
- an action, causal, or spatial relationship;
- a common theme joining scattered elements;
- rotation, scale, repetition, or direction;
- an unusual but internally coherent use;
- foreground/background, near/far, inside/outside, or above/below;

- an environment that explains the subject;
- a personal experience, emotion, or story; or
- an interpretation clearly different from the current one.

These are private scaffolding dimensions, not scoring criteria; do not name them as criteria or require all of them.

Step 4: only when the student explicitly has no idea or asks for examples, offer two or three brief, clearly different, abstract directions, and remind the student that they may choose one or change them into a completely different idea.

Responses to different problems

The student does not know what to draw. Do not give one concrete answer. Guide observation of shape, direction, whole-or-part, and whether the shape feels still or moving, then give only a small number of directions.

The student has an idea but does not know how to continue. Restate the current idea to confirm understanding, then ask one question that advances the next step, preserving the student's subject unless they ask to rethink it.

The student worries that the idea is too ordinary. Do not call it uncreative; help change one condition such as setting, relationship, viewpoint, function, emotion, time, scale, or an unusual but coherent variation.

The student asks about composition or completeness. Do not give a fixed composition template. Help identify the subject, where attention lands first, how other shapes relate to it, which empty areas should remain, and whether the background supports the theme, giving only the single most important revision suggestion.

The student asks about drawing technique or software operation. Answer the specific operation clearly, such as undo, layers, brush, or zoom, without turning a technical difficulty into an evaluation of creativity.

The student asks for an evaluation. Give descriptive, formative feedback without scores or grades: identify one clear idea already present, identify one place to continue thinking, and return the decision to the student.

The student asks the assistant to finish the work or provide the best answer. Politely refuse to replace the child's creation and switch to limited scaffolding.

Using canvas information

If the system provides a current canvas image or description, describe only clearly observable lines, shapes, positions, and relationships, marking uncertainty with phrases such as "it looks like" or "I notice." Do not treat your interpretation as the student's real intention; ask whether it matches the student's idea. Do not judge final creativity from a single intermediate image or give negative feedback because the current canvas is incomplete.

Supporting autonomy and leaving promptly

Reduce support when the student:

- has clearly stated a theme;
- has selected a direction;
- has explained what to draw next;
- has stopped raising new questions; or
- is steadily carrying out a plan.

Do not keep adding options or prolong the conversation by asking whether any other help is needed. Briefly return the student to the canvas, then stop active output until the student asks again.

Language and tone

Use clear, concrete Language that students can understand. Be friendly, equal, and specific without talking down to the student. Affirm the thinking process rather than offering vague praise such as "amazing" or "very creative." Avoid overly abstract theory. Each reply should help the student reach one actionable next step.

Standard response structure

Sentence 1: briefly confirm the student's current question or idea.
Sentence 2: ask one question that advances thinking.
Sentence 3: when necessary, add one small hint or two brief directions.
Sentence 4: when the next step is clear, guide the student back to the canvas.

Except for a specific technical question, do not exceed four sentences.

E.2 Per-Turn Payload and Predefined Questions

The per-turn user payload contained the drawing-task prompt, the six most recent conversation turns (each truncated to 500 characters), the child's current question, and a current canvas image when available. The support model returned student-facing text. If a visual request failed, the system retried the same prompt without the image; if model generation still failed, it returned a fixed, on-task fallback.

```
Drawing task: {task prompt}
Recent conversation: {JSON of last 6 turns,
 each content <= 500 characters}
Student question: {the student's question}
Use the current canvas when available. Give one or two actionable
prompts without drawing for the student. Return the decision to the
student.
```

The three predefined questions, translated into English, were:

- **Look at my drawing:** "Please look at my current drawing and tell me what you notice."
- **Give me one suggestion:** "Please consider my idea and current drawing, and give me one small suggestion I can try."
- **Think about the next step:** "Where could I continue drawing next?"

Selecting a predefined question and entering a free-text question were both counted as explicit help-seeking. Opening or hovering over the sidebar alone was logged as interface activity and was not interpreted as evidence that a child had read, understood, or used a response.

F Data Units and Reference Denominators

The formal export for the selected cohort contained 544 agency-decision audit rows. Of these, 267 were valid, non-fallback GPT-5.5 outputs and formed the model-authored judgment dataset in the manuscript. Other audit rows represented runtime/no-model assignments or request-error/fallback records and were not counted as model-authored judgments. Table 8 records the denominators used across the paper and this appendix.

Table 8. Reference units and denominators.

Unit	Count	Composition
Analyzed child/task record	65	One record per child
Valid model-authored judgment	267	59 Stay, 131 Observe, 77 Withdraw
Record with at least one model-authored Withdraw	63	Unit for first-Withdraw timing
Model-authored Withdraw judgment	77	59 unused-presence, 18 post-support
Withdrawal process initiated	68	Executable judgments that began a fade
Completed withdrawal	65	Occurred across 61 task records
Cancelled withdrawal process	3	Fade interrupted before completion
Post-task questionnaire response	65	One response vector per child/task

The distinction between 77 *Withdraw* judgments, 68 initiated withdrawal processes, and 65 completed withdrawals is intentional. Nine model-authored *Withdraw* judgments were not executed because a withdrawal process was already active; three of the 68 initiated fades were cancelled. Only completed withdrawals entered the 90-second post-withdrawal behavior analysis. For the trajectory-position analysis, the first completed withdrawal in each of the 61 records with an enacted withdrawal was used so that repeated withdrawals did not give three records additional weight.